

Review

Lithium-Ion Batteries on Board: A Review on Their Integration for Enabling the Energy Transition in Shipping Industry

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Abstract: The emission reductions mandated by International Maritime Regulations present an opportunity to implement full electric and hybrid vessels using large-scale battery energy storage systems (BESSs). Lithium-ion batteries (LIB), due to their high power and specific energy, which allows for scalability and adaptability to large transportation systems, are currently the most widely used electrochemical storage system. Hence, BESSs are the focus of this review proposing a comprehensive discussion on the commercial LIB chemistries that are currently available for marine applications and their potential role in ship services. This work outlines key elements that are necessary for designing a BESS for ships, including an overview of the regulatory framework for large-scale onboard LIB installations. The basic technical information about system integration has been summarized from various research projects, white papers, and test cases mentioned in available studies. The aim is to provide state-of-the-art information about the installation of BESSs on ships, in accordance with the latest applicable rules for ships. The goal of this study is to facilitate and promote the widespread use of batteries in the marine industry.

Keywords: lithium-ion batteries; ships; emission reduction; transport electrification; marine regulations; battery integration; BESS



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1. Introduction

The global demand to reduce CO₂ and other air pollutant emissions is affecting all industrial sectors, including maritime transport. Carbon dioxide, methane, sulphur/nitrogen oxides, and fine dust from ships are mainly emitted from fossil fuel combustion during propulsion and onboard electricity generation [1].

The Fourth International Maritime Organization (IMO) Green House Gas (GHG) Study of 2020 revealed that CO₂ emissions from ships reached 1.076 million in 2018, accounting for approximately 2.9% of the world's total amount. This indicates that GHG emissions from shipping activities are far from negligible. The IMO further highlighted that without additional measures, the CO₂ emissions from the shipping industry are projected to significantly exceed the planned objectives soon. However, in the same report is demonstrated that by implementing all proposed mitigation measures on ships built from 2025 onwards, a reduction in CO₂ emissions could be achieved by 2050 [2].

Practical actions for reducing shipping emissions can be categorized as technical, operational, and market-based measures (MBMs). The first two include measures oriented to the optimization of energy use or the adoption of alternative energy sources. MBMs, instead, are based on the payment of economical disincentives, proportionate to the amount of GHG emitted. This is the so-called "Emission Trading Scheme" (ETS) under discussion at IMO level and involving political and economic challenges still under discussion since 2009 [3,4]. Among technical actions, the electrification of propulsion and onboard ship systems is one way forward to meet emission requirements while improving overall ship

efficiency. For this reason, many modern ships are already electrically propelled with an integrated electrical power system (IEPS). New cruise ship designs are all fitted with an IEPS and many old ships have been retrofitted for this purpose [5,6]. In these ships, although thermal engines are still employed, they are used exclusively to drive synchronous generators necessary for electricity production [6]. Nevertheless, to achieve the net-zero carbon emission target and maximize energy efficiency, it is necessary to overcome the inefficient process to convert thermal combustion energy into electricity, and in this regard marine renewable energies in their different forms (wind, solar, wave, and tidal) can play a crucial role. Since these sources are often fluctuating and irregular, they lead to unacceptable power instability issues for the served grid, therefore their use is conditional on the combination with energy storage systems (ESSs) [7].

A comprehensive review of energy storage technologies for marine renewable energy sources (MRE) has been proposed by Wang et al. [8]. This work offers a perspective on where batteries stand among other energy storage systems such as pumped hydro storage (PHS), compressed air storage (CAES), hydrogen storage (HES), gravity storage (GES), and buoyancy energy storage (ByES). Focusing on battery energy storage systems (BESSs), Adeyemo and Tedeschi [9] offer a comparison of several technologies including lithium-ion batteries (LIBs) and supercapacitors. Moreover, this work includes a technology suitability assessment of peak shaving and spinning reserve services for offshore oil and gas platforms (OOGP), together with an introduction of the possible services for OOGP applications. Stampatori et al. [10] propose a review on LIB for transport decarbonization. They present market information divided for various applications and provide an analysis of the whole supply chain, including considerations on the effective environmental impact of different LIB chemistries. Verma and Kumar [11] compare performances of five battery types (lead-acid, lithium-ion, nickel-cadmium, silver-zinc and open water) used in maritime environment, discussing the global scenario of the marine battery market. In the work of Papanikolaou [12], the specific issues of high speed vessels are presented together with some indications on how to design and optimize battery installations for such types of ships. Kolodziejski and Michalska-Pozoga show ship test case projects with BESSs [13], proving that the electrification of ships is facilitated by certain types of ships (short-range ships, ferries, diesel-electric vessels, offshore dynamic units, dredgers, and tugs).

The present review provides evidence that, depending on the specific operative profile and mission type, all ship types can benefit from battery deployment. Analysing the track-records and press releases of recent new ship builds, it can be affirmed that lithium battery technology is the current commercial solution constituting the best compromise in terms of weight, space, performance, and cost [8,11,13]. For this reason, with the aim of limiting possible cases and hence enabling a detailed comparison that is more useful for real applications in the maritime business, this review is focused on LIBs.

Despite the increasing number of BESS commercial ship projects in the recent years, according to the best of the authors' knowledge, there is currently a limited number of specific studies on LIBs used on board ships. Therefore, thanks to a meticulous collection of information from electrical vehicles, terrestrial grid applications, and the study of the few existing marine reviews, the peculiarities of LIB ship applications have been identified in this review, with the purpose of underlining issues and opportunities that electrification is introducing in shipping. Some safety issues in the use of onboard LIB systems must be overcome; therefore, LIB use is subject to compliance with requirements and regulations imposed by international bodies and regulated by classification societies (Class).

The final objective of this work is to further extend the knowledge about the use of batteries in the marine industry compared to recent reviews [8,10–13]. In particular, herein a comprehensive explanation of the possible role of LIBs in the maritime industry has been proposed, depicting the elements needed to design a ship energy storage system, including an overview of the regulatory framework for big LIB installations on ships. After a discussion on possible ship services with relevant reference studies and projects from real applications, basic technical information about system integration has been summarized

to provide a state of the art about installation of BESS on board of ships according to last applicable rules for the shipping industry.

2. Use of Batteries on Ships

Batteries have already been in use on ships for a long time, with the main purpose being stand-by power for onboard general services or as an emergency energy source in case of the failure of the main power system. For over a century, lead-acid technology has been used, including as the main energy source for submarine propulsion [14]. Today, modern batteries for different vessel types can provide much more benefits to minimize the risk of blackouts, providing a faster response than emergency generators, stabilize the onboard micro grid during navigation, and provide stand-alone services when a ship is tied-up in port. More generally, the use of BESSs on ships can support all the ancillary services typical in marine electrical grids such as peak shaving, load levelling, power levelling, and power quality support [15,16]. The focus of this review is BESS applications for ship energy uses (see Figure 1) with a consumption higher than 0.5 MWh for typical daily ship operations [17].

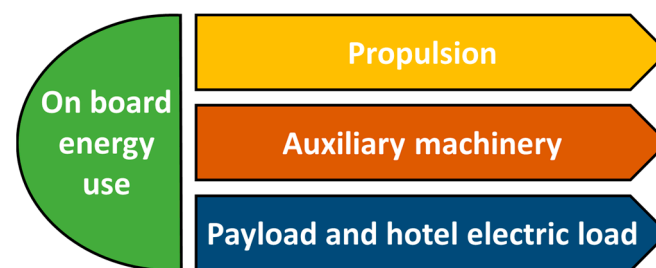


Figure 1. Primary functions having the greatest impact on energy use on board ships.

2.1. Possible Battery Services for Ships

Batteries are deployed on ships either as enabler for alternative energies sources or as energy buffer for propulsion and energy management actions. When BESSs are installed in place of combustion engines, a ship is defined as a “full electric” vessel, when instead, BESSs are installed in combination with a combustion engine, a ship is defined as a “hybrid” vessel [14].

In the first case, all services, including propulsion, are electrified and the ship’s power is supplied by electrical energy sources such as batteries or fuel cells [18]. In this configuration, one of the main benefits is the ability to maintain high propulsion efficiency for the ship’s entire operating range, while many of the vessels equipped with mechanical propulsion are designed for just one single operational point [19]. For this reason, all electric ships have a total efficiency, from energy storage to propeller thrust (tank to wake), of close to 70% (batteries used as an energy source), compared to the 30% of typical diesel–mechanical propulsion [20]. Therefore, electric propulsion can be considered the best solution in terms of energy efficiency and hence also in terms of emission reduction. Therefore, electric propulsion can be considered the best solution in terms of energy efficiency and hence also in terms of emission reduction. Furthermore, electric propulsion can be used for improving comfort and mitigating ship operation costs; indeed, electrical engines emit less noises and vibrations and hence require less maintenance [21]. However, an all-electric ship implies having a large amount of energy stored on board and, for many ship types, it is not easy to meet the global energy demand using the currently available technological solutions. Therefore, nowadays, there are limited applications of full electric vessels and, especially for medium and large-size vessels, most of the electric ships are hybrid, foreseeing electric operation modes and services [1] such as those summarized in Figure 2.

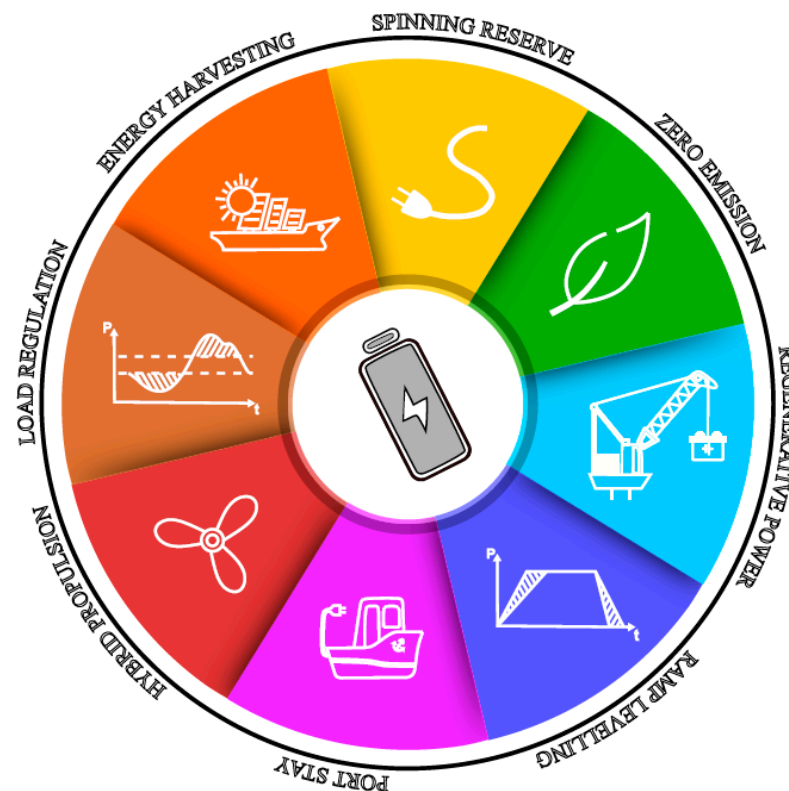


Figure 2. Ship-typical grid services and operational modes possible with BESSs.

Within hybrid ships, there are major differences between vessels with mechanical and electrical propulsion. In the case of the latter, since large diesel generators are adopted to produce a significant amount of electricity, the installation of a BESS can be easily integrated with the onboard electrical grid (see Section 4.3). When instead a ship has a conventional mechanical propulsion system, a solution for the adoption of batteries for ship services is to install a variable speed shaft generator (permanent magnet synchronous machine) coupled with the main shaft through a gear box [22]. In this configuration, as shown in Figure 3, the hybrid propulsion train is arranged with power take out (PTO) and power take in (PTI) systems. PTO allows electricity to be generated and stored in the battery when the propulsive load is low, while PTI helps deliver torque directly into the gearbox when it is needed to support propulsion.

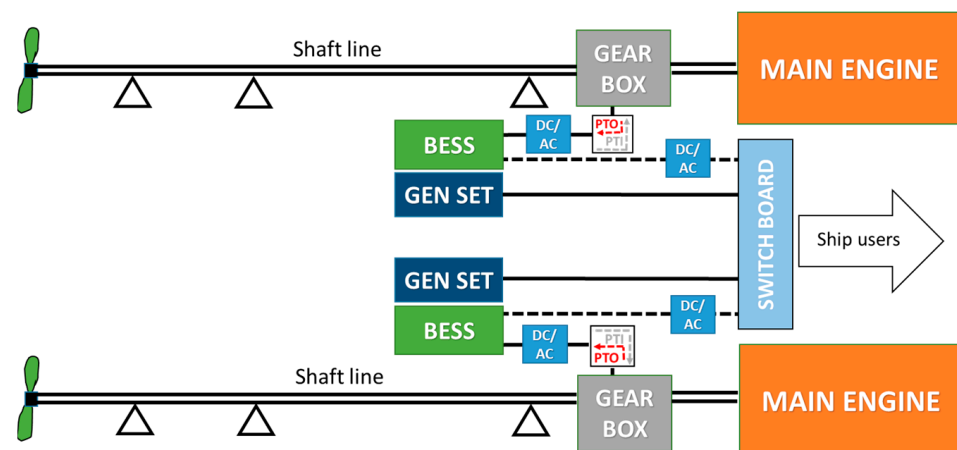


Figure 3. Possible arrangement of hybrid-mechanical propulsion with BESSs.

The **zero-emission** mode utilizes full electric propulsion for a limited time, reducing ship noise and vibration. Here, a BESS allows for switching-off all the onboard internal combustion engines. However, when fully powered by its own BESS, these ships are subject to specific technical limitations, such as max speed and maximum operation time, depending on battery power and energy content [23]. Another battery service provided by the onboard BESS is **port stay**. This is a cold ironing service where a ship's BESS can power all auxiliary and hotel loads (for limited time and power requests) without the need for internal combustion engines or a shore power connection [24,25]. Within services enabling the use of alternative energy sources are **energy harvesting and regenerative power**. The first service consists of a time shift of energy generation by storing energy from photovoltaic panels, and/or other renewable sources such as wind [26–28], and then releasing it when they are absent. Regenerative power improves ship energy management, using BESS to store energy obtained from the regenerative process of double speed engine braking (e.g., ships with cranes or large winches) [22]. The utilization of the accumulated energy can be for the rapid start of motors, peak levelling, or for other ship energy requests. Practical examples of regenerative power applications are ships with large winches such as tugs and anchor handling vessels or dynamic positioning vessels with drilling equipment with a power of up to 10 MW, or large thruster installations in the order of 1–6 MW with intermittent operations [5].

Some battery services foresee the use of BESSs in standby mode for obtaining cost savings in ship operation and maintenance [16]. In this context, a **spinning reserve** is used for safe manoeuvring with BESS acting as an energy backup in place of the need for one spare generator to be running during demanding ship manoeuvring operations as dynamic positioning operations (DP) [29,30].

During the operation of ships, there are frequent peak power loads due to ship speed changes during normal sailing and manoeuvres or as a result of rough sea conditions. In such situations, BESSs can operate to provide **peak shaving** services, with the batteries acting as an energy buffer and allowing for the installation of less power on board. BESSs deliver power when a ship's load request is higher than a threshold power level [31]. The battery units absorb the ship's load variations, and, in this situation, the engine will see a constant load. **Ramp levelling is used to enhance dynamic engine performance, providing instant power in place of generators or electric motors not having enough boost to perform the job** (e.g., alternative fuelled engines) [32–35].

Other services continuously use batteries with frequent battery charge and discharge cycles such as **load levelling** and grid stabilization. In load levelling, BESSs are used to keep the load of the diesel engine/generator constant, allowing them to operate at maximum efficiency by charging when the load demand is low, then contributing to the fulfilment of the load request [36,37]. In grid stabilization services, batteries are used for enabling ship smart grids by smoothing transients and reducing voltage and/or frequency deviations, hence providing grid regulation services [38]. A practical application can be combination with regenerative power storage services, where, due to the instantaneous power excess, it is necessary to guarantee grid stability by avoiding voltage drops.

2.2. Battery Use by Ship Type and Reference Projects

Today the majority of electric vessels are small leisure crafts; however, considering the total installed energy on board, battery installations for large ships play a significant role (e.g., ferries, offshore vessels, cruise ships, fishing vessels, etc.), though the number of large ships sailing remains low if compared with small crafts [39]. In general, the smaller is the vessel, the easier is to install batteries for the ship's power needs.

As summarized in Table 1, ship power requirements can vary enormously depending on ship type and application. Therefore, the electrification of ships is promoted due to several factors, such as propulsive plant type, onboard equipment, required power load, etc. For example, RoRo ships are a good candidate for electrification because, during the loading and unloading of passengers, cars and trucks offer high port times that are useful

for recharging operations. As proof, one of the first applications of a full electric ship is the ferry “Ampere” with an 18-car space on deck built in 2012.

Table 1. Different ship types with possible electrical services and suitable battery technology.

Ship Type	Power (MW)	Electrical Service	Battery Technology
 RoRo ferries	2–10	Load levelling Spinning reserve Full electric	ENERGY: High POWER: Service dependent CYCLES: Very high TYPE: NMC, LFP, LTO
 HSC ferries	1–15	Full electric Load levelling	ENERGY: High POWER: High CYCLES: Very high TYPE: NMC, LFP, LTO
 Cruises	20–70	Spinning reserve, immediate power Load levelling, grid stabilization Zero-emission short mission	ENERGY: Service dependant POWER: Medium high CYCLES: Medium high TYPE: NMC, LFP
 DP Class offshore vessels	30–50	Regenerative power Grid stabilization Spinning reserve	ENERGY: Service dependant POWER: Very high CYCLES: Service dependant TYPE: NMC, LFP, LTO, Supercapacitor
 Offshore Supply Vessels	10–20	Regenerative power Spinning reserve, peak levelling Grid stabilization	ENERGY: Medium low POWER: Very high CYCLES: Low TYPE: NMC, LFP, LTO
 Harbour tugs	2–4	Regenerative power Full electric Hybrid propulsion	ENERGY: High POWER: High CYCLES: Service dependant TYPE: NMC, LFP, LTO, Supercapacitor
 Fishing vessels	0.5–5	Load levelling, peak shaving Regenerative power Hybrid propulsion	ENERGY: Low POWER: Medium low CYCLES: Service dependant TYPE: NMC, LFP
 Shuttle tankers	10–20	Spinning reserve Peak shaving Load levelling	ENERGY: Medium POWER: Very high CYCLES: Low TYPE: NMC, LTO
 Various with cranes	5–15	Regenerative power Load levelling, peak shaving Zero-emission short mission	ENERGY: Low POWER: High CYCLES: High TYPE: NMC, LFP, LTO
 Yachts	1–30	Spinning reserve Peak shaving Load levelling Zero-emission short mission, port stay	ENERGY: High POWER: Low CYCLES: Low TYPE: NMC, LFP; LTO

Since the delivery of this vessel, other projects are exploring the possibility to have large battery packs that are able to guarantee full electric service also in medium and large-size ships. A more recent RoRo reference battery installation is the 117 m RoRoPax, vessel designed for the operator “Molslinjen” to be delivered in 2024. This ship foresees the installation of the “Enchandia” battery energy storage system (ESS) with a total capacity of 7000 kWh. Other types of ships, such as high-speed craft (HSC) ferries, allow for adopting

full electric solutions due to their relatively low energy requirements. There are already many seagoing vessels such as the “MS Medstraum”, a 30 m HSC vessel delivered in 2022 and part of the EU-funded project “TrAM”. This vessel has a “Corvus Dolphin Power” battery ESS with a total capacity of 1524 kWh providing all electric propulsion at a service speed of 23 knots.

Some special ships have the potential for regenerative power. This is the case in sea-going cargo carriers with cranes (e.g., containers, tankers, bulk carriers) benefiting from energy regeneration during the lowering of cargo. The “Kryssholm” a 90 m vessel retrofitted in 2021, is a bulk carrier equipped with a 254 kWh “Corvus Orca” battery ESS, designed for regenerative power, load levelling, peak shaving, and zero-emission mode during port operations. Other special ships, such as drilling vessels, have complex electric grids hosting several large users; hence, they are subject to fluctuations and batteries can be adopted for grid stabilization services. An example of this is the “North Sea Giant”, a 153 m DP3 ship retrofitted in 2018 with a 2034 kWh “Corvus Orca” battery ESS. Offshore supply vessels (OSVs) are instead characterized by a propulsion system that is composed with two main shafts or azimuthal thrusters, combined with multiple bow and stern thrusters that are necessary for DP capabilities. On these ships, the installation of batteries is promoted by the high redundancy required by the Class that a BESS can provide through spinning reserve services as per the “Norwind Breeze”, an 82 m offshore support vessel delivered in 2022 with a 496 kWh “Corvus Orca” battery ESS.

In addition to the power requirements that depend on ship type, there are big differences that depend on the electrical services that are offered. Thus, different ship types’ operational profiles and specific ship missions may require different BESS designs in terms of technology, chemistry, and technical requirements (See Table 1, the “Battery type” column). Therefore, the power and energy involved may change for different operating modes. This is the case for harbour tug ships that, despite having highly variable mission types, can also be good candidates for lithium-ion battery installation and hybrid systems that are often used for tugs in order to combine high power and high specific energy in a single BESS [40]. For several tugs, the operational profile foresees extended periods at low power, with a low ship speed or being moored in port, and short missions with high power requests. A reference project is the “HaiSea Wamis”, a 28 m full electric tug delivered in 2022 and featuring a 5288 kWh “Corvus Orca” battery ESS.

Among different BESS technologies, lithium-ion batteries can effectively meet various demands on marine vessels for each of the above scenarios. However, it is crucial to consider how battery performance affects their behaviour in different operational services. Charging and discharging rate capabilities, depth of discharge (DoD), storable energy, and lifetime energy throughput significantly impact lithium-ion battery behaviour on marine vessels during peak power demand, long-duration cruising, and when docked [37]. Therefore, these factors should be carefully managed to optimize both vessel operation and battery performance and safety. For instance, during peak power demand, high current rates can generate excessive heat, accelerating degradation and increasing thermal runaway risk. Furthermore, lifetime energy throughput, which refers to the cumulative amount of energy that can be drained from the battery before it becomes unsuitable for its intended use, can lead to an inability to meet power demand as a result of capacity loss and elevated internal resistance. For long-duration cruising, a typical low rate of discharge minimizes the stress on batteries, extending their lifetime. However, if the energy that can be stored in a BESS is limited, the occurrence of frequent deep discharges may lead to a shorter battery life. In this scenario lifetime energy throughput has a strong impact since a reduced battery capacity affects the navigation distance and speed of the vessel. Maintaining a moderate depth of discharge (at maximum 80%) is an effective strategy to balance performance and longevity. When docked, the charging rate is the main stressing factor of the batteries. However, when slower charging is practicable by prolonging port stay periods, battery life can be extended.

In summary, although it has been amply demonstrated that batteries can meet the operating requirements for several types of ships, optimal system sizing and the correct selection of battery technology based on the possible operating scenarios are necessary to ensure their efficient and durable use [9,41].

3. Lithium-Ion Batteries for the Marine Industry and Future Perspectives

Among mature electrochemical storage technologies, whose effectiveness and reliability have been widely proven, there are established chemistries such as lead-acid, nickel-based, lithium-ion, high-temperature sodium, redox flow, and supercapacitors [7]. However, lithium-ion batteries (LIB) currently represent the most suitable technology to meet the technical performance, size constraint, and reliability requirements for the maritime industry [11]. The first commercial devices with a lithium metal anode date back to the 1970s, while Sony (Tokyo, Japan) initiated their market penetration in the 1990s. Sony proposed a battery with improved safety and durability thanks to the use of an intercalating graphite anode [42]. Despite their relative youth, lithium-ion batteries are today the most widely used electrochemical storage system. This popularity is due to their high specific power (up to 2000 W/kg [43,44]) and specific energy (100–250 Wh/kg as per Table 2), which allows for scalability and adaptability from small portable devices to electric vehicles and large stationary systems. However, their main limitation continues to be their cost, which is less competitive than that of other technologies. It is worth noting that the manufacturing capacity of LIBs for mobility and stationary energy storage applications was about 150 GWh/y in 2018 and 700 GWh/y in 2022 [45], and the global annual battery demand is predicted to reach about 1700 GWh by 2025 and 4700 GWh by 2030 [46]. Therefore, a specialized economic study on the cost of batteries shows that the production cost of LIBs will halve in 2030 if compared with 2018 [47], reaching the price of 100 \$/kWh before 2030 for NMC automotive batteries [48].

3.1. Lithium-Ion Battery Types

Considering the variety of anode and cathode chemistries, different LIBs can be identified. Usually, their name is indicative of the cathode composition, i.e., lithium-iron-phosphate (LFP), lithium-manganese oxides (LMO), lithium-cobalt oxides (LCO), and mixed nickel-cobalt-manganese/aluminium oxides (NMC and NCA). On the other hand, when lithium titanate (LTO) is used instead of graphite, they are also identified according to the anode chemistry. A list of the main types of LIBs and their key features can be found in Table 2 where in the column “marine applications”, only the chemistries deemed more suitable for ship applications have been commented.

For more than 30 years, lithium cobalt oxide “LCO”, with the formulation LiCoO_2 , was the dominating cathode material used in commercial lithium-ion batteries due to its high voltage level and specific energy [49]. However, this cathode is affected by some limitations related to the high cost of cobalt, low thermal stability, and the loss of capacity with deep charge–discharge cycles [50,51]. In fact, there are safety concerns related to oxygen release at high temperatures giving possible thermal runaway issues with consequent high flammability and the risk of explosions. Moreover, LCO batteries, when subject to deep charges, may lead to excessive lithium removal from the structure, causing lattice distortion and performance degradation within a relatively short number of cycles [52].

Table 2. Comparison of different LIB types and possible uses in marine applications.

Chemistry	Cycles Number	Cell Voltage (V)	Specific Capacity (Ah/kg)	Volumetric Capacity (Ah/L)	Specific Energy (Wh/kg)	Main Features	Uses IN Marine Applications
LCO	500–1000 [53]	3.8 [54] 3.9 [55]	145 [54] 130–155 [55]	550 [54]	150–190 [53]	High specific energy Low thermal stability Limited safety and power Low cycle life Toxicity and High cost due to Cobalt	-
LMO	1000–1500 [53]	4.1 [54] 4 [55] 3.7 [56]	120 [54] 100–120 [55]	596 [54]	100–140 [53]	Average number of cycles High C-rates Low specific energy Low toxicity Low cost (Cobalt free)	LMO used in combination with LTO anodes
NMC	1000–2000 [53]	3.7 [54]	170 [54] 160–170 [57]	600 [54]	140–200 [53]	High voltage High specific energy Rapid aging in terms of specific energy fade Good thermal stability Tailored composition possible	Largely adopted, thanks to flexible features in terms of energy and power capabilities
NCA	1000–1500 [53]	3.7 [54]	200 [54]	700 [54]	200–250 [53]	High specific energy Good calendar life Low cycle life Low safety Moderate cost due to limited Co	Used for application with high energy request
LFP	up to 2000 [53] 1000–2500 [55]	3.4 [54] 3.3 [55]	165 [54] 150–170 [55]	589 [54]	90–140 [53]	High safety High number of cycles Low specific energy High self-discharge Wide SoC window (15–100%) “Eco-friendly” materials Low cost for Co absence	Used for applications due to the best safety and flat voltage curve
LTO	above 3000 [58]	2.6 with LMO [59]	175 [54]	77 with LFP cathode [59]	50–80 [60]	High cycle life High safety High C-rates Low specific energy	Used for applications thanks to fast charging capability, high power and very high cycling

Lithium manganese oxide “LMO”, with the spinel structure LiMn_2O_4 , is the economically valid alternative to the LCO cathode due to its advantages of high thermal stability, safety, low cost, environmental friendliness, good specific power, and acceptable specific energy [61]. However, LMO has a limited resistance to continuous charge and discharge cycles due to phase transformation phenomena [62]. In particular, layered LiMnO_2 structures are subject to partial transformation into spinel during the extraction of the Li ions with consequent voltage and capacity fade [63]. In addition, the LMO structure has the tendency to dissolve in the electrolyte, and the free Mn ions destabilize the protective passivation layer (SEI) formed at the anode [64]. Both cathode structural changes and anode impedance increase are two relevant effects that are the cause of cell aging [65,66].

An alternative cathode material, less expensive than LCO but having higher capacity than LMO, is lithium nickel-manganese oxide “LNMO” with the most common formulation $\text{Li}(\text{Ni}_{0.5}\text{Mn}_{0.5})\text{O}_2$ or the Ni-rich structure $\text{LiMn}_{1.5}\text{Ni}_{0.5}\text{O}_4$ [67]. An LNMO-based cathode is attractive because its good theoretical specific capacity in combination with a high working voltage leads to a 20% higher specific energy than an LCO cathode, while reducing costs thanks to the absence of expensive cobalt. Moreover, thanks to the high voltage, the LNMO cathode can be effectively coupled with the higher voltage anodes, such as LTO, to obtain a high specific energy cell [68,69]. Additionally, the high Li ion diffusivity through the lattice of the spinel structure of this cathode enables the high C-rate performance of the cell [56,70]. Currently, Ni-rich cathode materials represent the main choice for “NCA” commercial power batteries due to their high specific capacity and cost advantages compared to LCO [71]. In fact, the lithium nickel oxide cathode, in its simplest layered LiNiO_2 form (LNO), has a crystalline structure similar to the LCO cathode with comparable specific capacity. Therefore, the performance, in terms of capacity and stored energy, is comparable for these two cathode materials with LNO being more convenient due to the lower material cost obtained thanks to the substitution of cobalt with nickel. However, LNO cathodes exhibit chemical and thermal instability, which restricts

their use in their pure form. Doping with cobalt (Co) proves to be an effective solution for reducing lattice instability. Additionally, the incorporation of other dopants, such as manganese (Mn) and aluminium (Al), is commonly employed to enhance both thermal stability and electrochemical performance [72]. It is for these reasons that mixed metal oxides are the used cathode materials instead of their pure counterparts. Among these, the lithium nickel cobalt aluminium oxide cathode (NCA), with the most common formulation $\text{LiNi}_{0.85}\text{Co}_{0.1}\text{Al}_{0.05}\text{O}_2$, was adopted for commercial use as a valuable alternative to the LMO cathode due to its improved cycle stability. Furthermore, NCA cathodes compared with LCO are less expensive thanks to the sufficiently high theoretical specific capacity and reduced quantity of cobalt [73]. However, it is reported that NCA cathode degradation is strongly dependent upon temperature and depth of discharge (DoD), due to micro-crack generation at micro-grains resulting in fast capacity fade and increase of the cell resistance at high temperature [74].

Another widely distributed commercial mixed metal oxide-based cathode material is lithium manganese nickel oxide “NMC”. One of the most typical formulations is $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ (NMC111), which can be considered a part of the layered LMO cathode family, where Mn ions are partially replaced by Co and Ni ions to stabilize the structure [75]. Although NMC theoretically has the highest electrochemical potential, this capability is often constrained by the type of electrolyte that is used. Oxidation within the electrolyte can lead to oxygen loss from the cathode and the dissolution of transition metals from its structure [76]. A combination of the high specific energy of nickel-rich material and cobalt’s/manganese’s capability to stabilize the structure are the characteristics that mix together to tune cell performance in the realization of different cathode compositions (NMC 811, NMC 622, NMC 532) [77].

Today, NMC and NCA cathodes have replaced the LCO cathode due to the fact that they are more economical due to the lower amount of cobalt in the formulation; furthermore, both electrodes have a higher number of cycles and better safety due to their higher thermal stability compared with LCO.

Lithium iron phosphate, with the typical formula LiFePO_4 “LFP”, is the most common polyanion compound that is used for the fabrication of cathode electrodes in commercial cells. This material has an olivine structure with a strong covalent PO_4 bond that reduces oxygen release as the temperature increases, improving the thermal stability of the cathode [77]. For this reason, LFP cathodes have reduced risks of explosion and low flammability even in the event of over-charging or short circuiting. The high safety, together with its acceptable specific capacity, flat voltage plateau, abundant and low cost raw materials, and good environmental compatibility have made LiFePO_4 the preferred cathode material for commercial lithium-ion cells when medium specific energy values are acceptable [78,79]. The main disadvantages of LFP cathodes are the relatively low voltage that results in lower specific energy when compared with other cathodes [80]. Another limit is their low electronic conductivity and the slow lithium ion diffusion during both the intercalation and de-intercalation processes, which can limit the maximum C-rate of the cell. However, to enhance electronic conductivity and improve the overall electrochemical performance of the LFP cathode, several strategies are commonly employed. These include reducing the particle size of the cathode, incorporating carbon into the positive electrode, and utilizing anionic/cationic doping [81]. Thanks to these actions, it is possible to reduce the resistance of the material, favouring the release of high currents and, therefore, higher power.

Despite most lithium-ion batteries taking their name from one of the cathode materials presented above, a type of cell takes its name from the anode. This is the case of LIBs, which cells use lithium titanate “LTO” as the anode. The main advantages of this anode material are its excellent thermal stability [82] and extremely high cycle life [58]. Both advantages are attributed to the almost zero-strain behaviour of the material during lithium ion insertion/extraction, which limits the volume change to be lower than 0.4% [83]. This feature has the beneficial effect of limiting the mechanical ageing of the LTO electrode

and, at the same time, avoiding the cracking of the protective SEI layer during cycling, helping to prevent the reaction of the electrolyte with the anode and the related safety concerns [84]. Another benefit associated with the voltage level of LTO is its ability to prevent the deposition of metallic lithium, which could lead to dendrite formation and could induce internal cell short circuits [85]. The main disadvantage of the LTO anode is its high electrochemical potential (1.55 V vs. Li/L⁺) which results in a lower cell voltage coupled with typical cathode materials. Poor specific energy due to the low cell voltage together with the current excessive cost are the limiting factors of this technology [86].

3.2. Lithium-Ion Battery Ageing

As for other energy storage technologies, LIBs are subjected to degradation due to ageing during their life. While the calendar life is associated with the gradual capacity loss over time, regardless of usage, the cycle life pertains to the capacity reduction resulting from the number of charge–discharge cycles during active use. In both cases, ageing is caused by alterations in the chemistry of the electrodes and the electrolyte as a result of operational parameters and external causes which result in different kinds of degradation phenomena [87]. In particular, calendar aging is affected by ambient temperature, humidity and the storage state of charge (SoC), which also induce degradation during the rest periods of the battery. On the other hand, the type of usage mainly determines cycle aging.

A schematic representation of the correlation between the different causes of ageing, the mechanisms, degradation modes, and their final effect on LIB performance can be found in Figure 4. Among the most common operating parameters affecting the degradation of LIBs, there is current load, which, in general, accelerates ageing. The depth of discharge, depth of charge, and the average SoC required by the operative profiles can also significantly vary the performance in terms of cycles by imposing more degradative full charge–discharge cycles or more conservative intermediate SoC operation [88]. High-temperature operation is among the most relevant ageing accelerators [89]. The optimal operational range for a LIB is 15–35 °C [90], therefore a control system, able to maintain the temperature in the proper range, is essential for prolonged battery life. Furthermore, mechanical stress induced by manufacturing, the normal packing compression of cells in battery modules, or accidents can also contribute to directly alter the performance of LIBs through structural and electrochemical changes inside the cells [91]. All the above causes lead to different degradation mechanisms of the active components of the cells. Among these, the most relevant are electrolyte decomposition leading to solid electrolyte interphase (SEI) growth at the anode and cathode; particles cracking as a result of volume changes during cycling; lithium plating; transition metal dissolution from the cathode; solvent co-intercalation; binder decomposition; and current collector dissolution. All these degradation phenomena are categorized in three main degradation modes being loss of lithium inventory (LLI), loss of active material (LAM), and conductivity loss (CL) [92]. The final effects of the previous degradation modes are manifested as capacity fade which limits the lifetime of the battery, and as power fade due to an increase in internal resistance.

3.3. Lithium-Ion Battery Hazards

As is known, LIBs present specific and significant hazard potential due to their high stored energy and due to the presence of reactive materials and unstable components inside the cells. As schematized in Figure 5, the three main significant interconnected hazards associated with the LIBs are the electrical hazard, the chemical hazard, and the fire–explosion hazard.

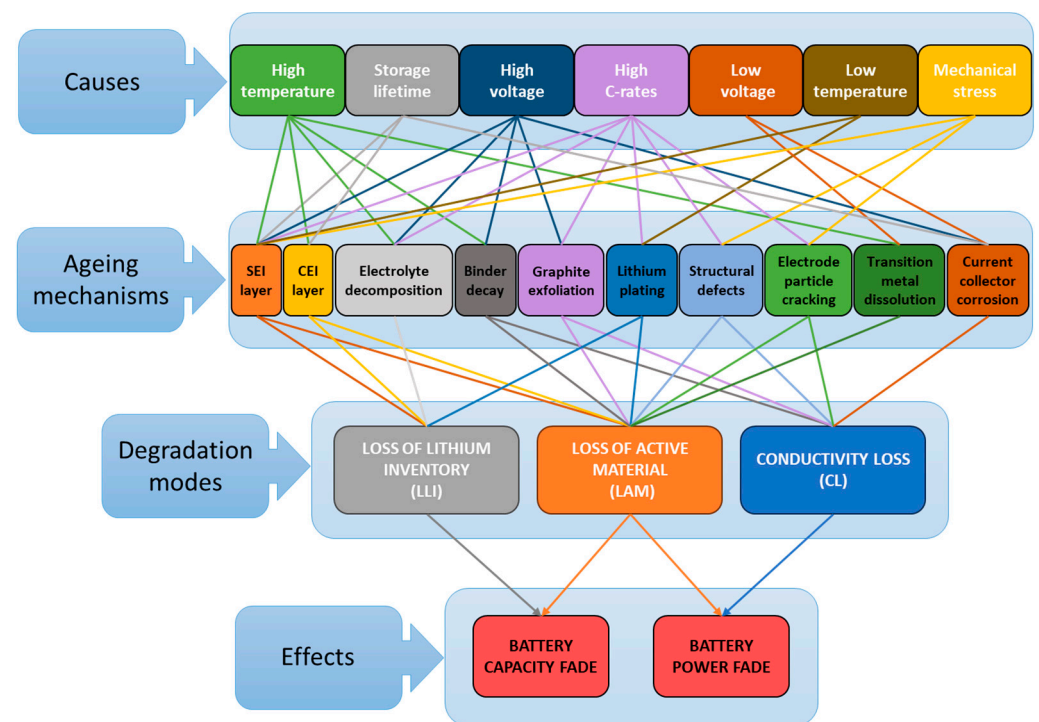


Figure 4. Correlation between causes, ageing mechanisms, and macroscopic effects in lithium-ion batteries.

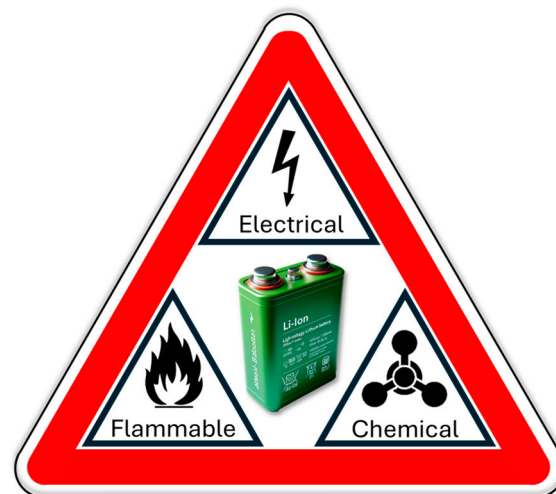


Figure 5. Main hazards associated with LIB use, storage, and transportation.

The electrical hazard is less relevant when single lithium-ion cells are considered, since their maximum voltage never exceeds 4.2 V, while it is of primary importance for ships where many cells are connected in series in modules and systems. On the other hand, chemical, and fire–explosion hazards also pertain to the use of single cells. The LIB chemical risk is associated with the exposure to the internal reagents of the cells, which include reactive salts, volatile organic compounds, metals, and metal oxide particles. Although lithium cells are hermetically sealed in normal conditions, they may release their internal content due to the degradation of their parts or as a consequence of accidents such as mechanical damage or explosive events [93]. If the release of these substances occurs in an enclosed space, such as in a ship compartment that is used for energy storage, it can result in the formation of a toxic atmosphere with concentrations that exceed exposure limits [94]. There are various hazardous compounds which can be released by LIBs or

generated after their interaction with the atmosphere; however, most of them come from the electrolyte [95]. For instance, lithium hexafluorophosphate salt (LiPF_6) in mixed organic solvents is typically used as the liquid electrolyte in most lithium-ion cells. These two substances under conditions of abuse can cause a substantial increase in cell temperature and, under these conditions, react producing phosphorus oxyfluoride gas (POF_3). Once released into the atmosphere, POF_3 rapidly reacts with humid air hydrolysing to produce hydrofluoric acid (HF), a well-known corrosive and toxic compound [96]. Many other toxic compounds can be released from the decomposition of the LIB electrolyte after fire events. These include flammable and toxic gaseous mixtures (H_2 , CO, CO_2 , CH_4 , C_2H_4 , C_2H_6 , C_3H_8 , HF, POF_3 , PF_5 , ethyl fluoride, propylene, etc.) [97] or the same volatile organic solvents [94]. In addition to the toxicity of substances released due to the electrolyte, metal residues from the decomposition of the cathode material can be a source of exposure risk. As stated above, nickel and cobalt-based lithium-ion cells, such as NMC and NCA, are the most diffuse cathode chemistries. However, nickel and cobalt compounds are classified as carcinogens, and they are known to provoke respiratory issues if inhaled and allergic reactions when in contact with the skin. Other elements constituting the cathode composition, such as aluminium and manganese compounds, can also result in toxic effects after inhalation exposure [98].

LIB fires pose hazards that are significantly different to other fire hazards in terms of initiation, spread, duration, toxicity, and extinguishing. Ignition is promoted by the solvent used in the electrolyte. This is a mixture of carbonates such as dimethyl carbonate (DMC), diethyl carbonate (DEC), ethyl methyl carbonate (EMC), ethylene carbonate (EC), or propylene carbonate (PC). All these carbonates are highly flammable and, due to their high volatility, can form explosive mixtures with oxygen once released into the atmosphere. Electrolyte decomposition can also produce many other flammable gasses, including hydrogen, methane, ethane, carbon monoxide, propene, etc., which contribute to the ignition of a fire after they vent from the cell [99]. Cathode electrode materials that are made of metal oxides or oxygen-containing compounds can also cause rapid fire ignition and propagation. The main risk is their thermal decomposition once the internal temperature of the cell is increased, which leads to the release of a large amount of oxygen from the crystal structure. This acts as an oxidizing agent for the flammable electrolyte/combustible gases and could lead to an explosion risk due to a pressure increase inside the cell [100]. Thermal runaway is the typical mechanism leading to fire or explosion in LIBs. This phenomenon represents a thermal instability triggered by internal or external factors, which then sustains itself through uncontrolled chemical reactions involving various cell components. The mechanism behind thermal runaway is often linked to the degradation of the passivation layer (SEI), primarily covering the surface of the negative graphite electrodes. As a result, carbonaceous species are released into the electrolyte [101]. In fact, when the temperature inside the cell exceeds $80\text{ }^\circ\text{C}$, the solid electrolyte interface (SEI) begins to progressively dissolve in the electrolyte. This dissolution triggers exothermic reactions, releasing heat. As the temperature continues to increase, the electrodes, separator, and other cell components become involved in the exothermic process. This rapid acceleration could lead to thermal runaway with possible gas release, fire, or even battery explosion.

Thermal instability phenomena may have several causes: internal short circuits; overcharging; excessively high operating temperatures; overheating from external sources; mechanical damage; battery management system failures; and manufacturing issues [93]. Therefore, this type of problem requires special protection systems to be designed case-by-case depending on the selected chemistry. Indeed, different cell compositions may have different ignition points and fire propagation modes. For example, LTO cells have a graphite anode with lithium-titanate that removes the issues related to carbon reactions with electrolytes hence reducing the fire and thermal runaway risk. Conventional LCO cells, instead, exhibit a pronounced tendency toward thermal runaway due to the decomposition of the cathode material already at temperatures slightly above $150\text{ }^\circ\text{C}$. However, the utilization of mixed metal oxides, as seen in NMC cells, enhances thermal stability by

raising the cathode decomposition temperature to a range of 220–280 °C [102,103]. LFP is the safest cathode material among the diverse options. It exhibits excellent thermal stability, remaining stable even at temperatures exceeding 250 °C, and it releases minimal oxygen during decomposition [104].

3.4. Battery Perspectives in the Marine Industry

Current battery research is focused on new inexpensive materials and new technologies to obtain LIBs with better performance in terms of specific energy, specific power, and safety. This is challenging because less expensive materials can give high energies, but they have low power and inferior aging capabilities, together with lower thermal stability and, consequently, inferior safety. Looking into the future, some promising concepts, such as solid-state batteries and lithium–air batteries, are under development [11].

Solid-state batteries use a solid ceramic electrolyte as a separator, which allows the use of metal lithium instead of a typical intercalation graphite anode. Significant benefits of this technology are the higher achievable specific energy than standard lithium batteries and the safety due to the obstacle for dendrite formation together with the non-flammable nature of the solid electrolyte. However, at the current state, its weaknesses are related to the low conductivity of the solid electrolyte, which leads to a limited specific power capability and high production costs [105]. If this battery will enter the market, according to the available theoretical data, future marine battery packs might be available in about 10 years, and the specific energy content could be triple if compared with lithium-ion batteries, with consequent similar increases in achievable ship ranges [106].

Lithium–air batteries, using oxygen from air at the cathode, are a promising technology due to a very high specific theoretical energy (e.g., 3400 Wh/kg) [107] and promising safety performance when solid state electrolytes is used. However, many critical issues have to be addressed before the practical use of this technology, such as the low round-trip energy efficiency, low specific power, use of pure air, and limited deep cycling [108]. According to current developments, this type of battery could be available for market applications in about 15–20 years and, theoretically, it promises specific energy content that will be four times the current energy content of conventional lithium-ion technology [11].

Beyond lithium-based batteries, sodium-ion is an emerging battery technology with promising potential as an energy storage system for marine transportation [109]. Similar to LIBs, sodium-ion batteries (SIBs) comprise an anode, a cathode and the electrolyte, but they differ in their active material, utilizing the more abundant, environmentally friendly, and cost-effective sodium element instead of lithium. Therefore, they offer some benefits over lithium technology in terms of cost, safety, and sustainability. Although SIBs have already been introduced into the market, they are still a new technology, and some challenges need to be addressed before they can be widely adopted in marine transportation. One of these is the necessity to increase their specific energy. However, the typical specific energy of SIBs is close to 120–140 Wh/kg per cell, which is on a par with LFP LIBs [110]. This means that SIBs need larger sizes or heavier weights to achieve the same amount of energy or power output if compared with their more performant sister technology based on lithium ions. Therefore, research and development are still needed to make sodium-ion batteries a viable option for marine transportation in the future. A recent development in sodium-ion battery technology is the use of sea salt, or NaCl, as a raw material for cathode synthesis. This can further reduce the production cost and mitigate the already low environmental impact of sodium-ion batteries by using abundant and cheap salt. In addition, sea salt can also improve the performance of sodium-ion batteries by providing sodium carbonate (Na_2CO_3) or sodium chloride (NaCl) ions to the cathode material [111]. A special class of SIBs, more appealing for marine applications, is the rechargeable seawater battery. The first not-rechargeable primary seawater batteries were developed in the 1940s and mainly used in military applications. Recent research is re-energizing this energy storage system by proposing rechargeable seawater-activated batteries as a promising technology for marine applications [112]. A rechargeable seawater battery consists of an anode and a seawater

cathode as an unlimited source of Na^+ ions. The charge is stored as an elemental sodium metal in the anode, which is protected from the seawater cathode by a Na^+ ion-conductive solid electrolyte, able to block water molecules while physically separating the anode from the cathode. The seawater cathode is usually exposed to the ambient air to involve dissolved oxygen species (DO) and the oxygen evolution reaction (OER) during the charge and discharge processes. The full cell reaction ($4\text{Na} + \text{O}_2 + 2\text{H}_2\text{O} \leftrightarrow 4\text{NaOH}$) produces a theoretical voltage of 3.48 V vs. Na/Na^+ [113]. If the cathode directly contacts seawater, the abundant sodium ions can migrate into the anode during the charging process leading to a huge specific capacity limited only by the volume of the anode compartment. The direct use of seawater could make these batteries especially suitable as energy storage systems for the marine industry, including ships, offshore, and seaside power sources.

4. Battery System Integration for Ships

Among transportation battery applications, BESSs for ships are currently some of the bigger installations in terms of size. A conventional BESS for an electric car has about 60 kWh of energy (Tesla Model 3), a long-haul truck can reach up to 1000 kWh [114], and a small ferry commuter ship requires 4300 kWh [115]. The energy required by the RoRo ferry projects under development is up to 70 MWh, as exemplified by the “Stena Elektra” project. Therefore, the order of magnitude of the energy content is different when comparing marine and road transport applications. Thus, although the adopted battery technology is usually the same for different transport applications, there are peculiar issues to overcome for successful integration on ships. These are related to the large energy scale, but also to ship-specific constraints, such as onboard space availability and weight limitations necessary to ensure the ship’s buoyancy conditions and stability requirements. Hence, the process of integrating batteries on board ships necessitates not only ensuring sufficient battery capacity to meet the ship’s specific operational requirements, but also adhering to weight and space constraints, fulfilling Class requirements, and ensuring economic viability (see Figure 6). This process includes facing several technical challenges related to harsh marine conditions such as salt corrosion, vibration, ventilation, humidity and pressure control, and a wide operating temperature range, including high and low extremes [116]. Therefore, before proceeding with the integration of batteries on board a ship, the selection of the appropriate chemistry and the definition of a specific system configuration are necessary.

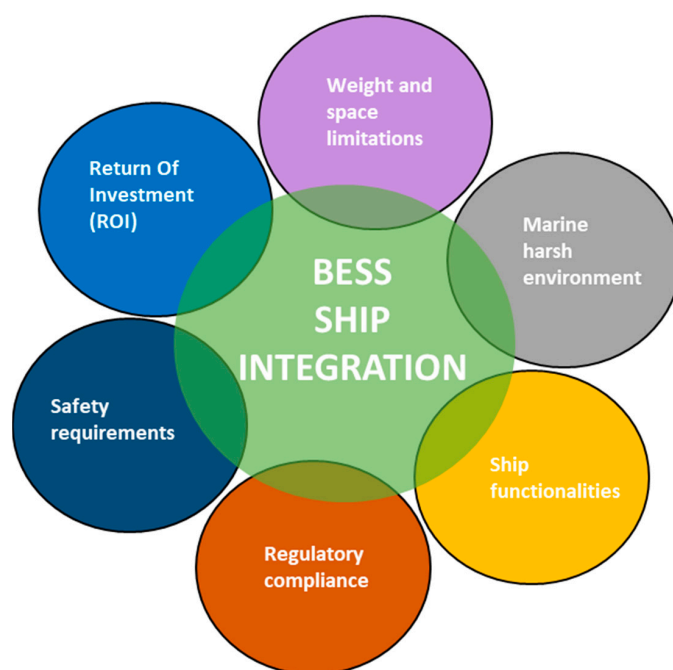


Figure 6. Challenges faced during the battery energy storage system ship integration process.

4.1. Battery Selection Criteria for Ship Applications

Battery performance depends on the chemical composition, technology, and system arrangement. The design process can be divided into two phases, cell selection and system selection. Selecting the appropriate cell chemistry requires a trade-off between capacity, voltage range, maximum power, and aging performance.

One of the main parameters by which to judge a battery for transport applications is the specific energy, i.e., the energy content linked to its dimensions (Wh/m^3 or Wh/L) and weight (Wh/kg). Figure 7 shows the average specific energy values for various rechargeable battery cell technologies. Here, for various battery technologies, it is possible to observe specific energy ranging from 20 Wh/kg to 200 Wh/kg . Due to their low specific energy, lead-acid batteries would be the worst choice for marine applications where a high energy storage capacity is required. On the contrary, lithium batteries, and, in particular, some specific chemistries, (see Section 3.1) make possible to minimize weight and size for the required energy.

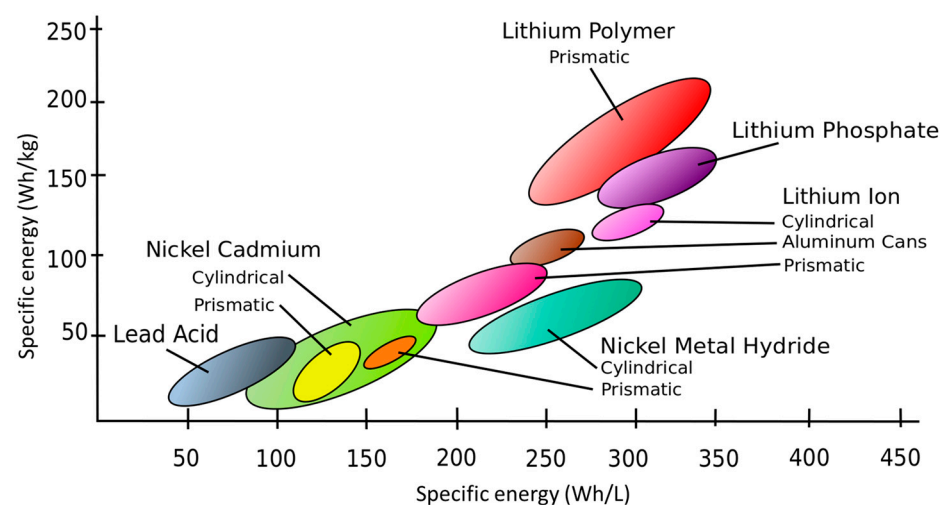


Figure 7. Specific energy for different types of secondary batteries: mass energy density (Wh/kg) vs. volumetric energy density (Wh/L). Licensed under CC BY-SA 3.0, Author Barrie Lawson.

Specific power can be expressed as mass power density (W/kg) or volumetric power density (W/L) and it is also used to compare different batteries [117,118]. A Ragone plot (Figure 8) helps select which technology is the most suitable for a certain application. As can be seen in Figure 8, typical battery storage systems show a specific power of between 10 and 10,000 W/kg and, especially for LIBs, it can be observed that a high specific power corresponds to a relatively low specific energy. In fact, fast charging processes require the minimum possible resistance, which is achieved with electrodes of small thickness and large active surface area, contrary to what is necessary to have a high-energy capacity. Therefore, to obtain the best combination, it is necessary to find the right balance between energy and power. One solution could be the use of multiple small cells, capable of providing high power, connected in parallel to achieve the desired energy capacity, rather than a single cell with a high capacity but incapable of providing high power. High power values are necessary when fast charging is required or when it is necessary to compensate for peak power loads. In applications where there are very high power requests, hybrid storage systems can be deployed, combining batteries with supercapacitors having a specific power of over 20,000 W/kg [119].

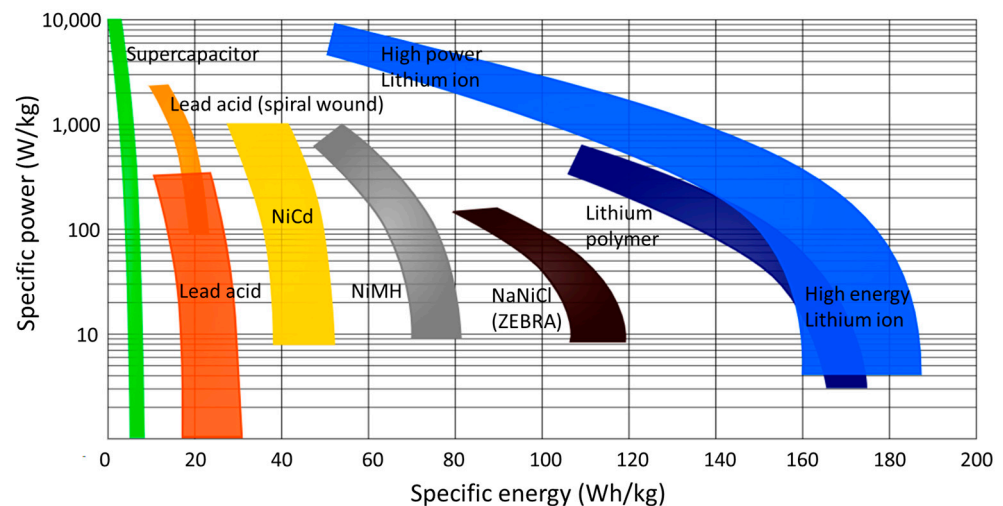


Figure 8. Ragone plot showing specific power vs. specific energy for different energy storage technologies. Licensed under CC BY-SA 4.0, Author FelixFlIX. Data from Überblick über die Speichertechnologien, Dirk Uwe Sauer at the Wayback Machine.

Currently, among the lithium-ion batteries, the most suitable cell chemistries for marine applications are NMC, NCA, LFP, and LTO, which satisfy the technical requirements necessary to meet typical ship operating profiles. For this reason, these are often used within marine research projects, for example in the recent “TecBIA project”, which is aimed at creating a prototype vessel equipped with a hybrid propulsion system composed of diesel generators, fuel cells, and lithium-ion batteries. Identifying the most suitable chemistry for BESSs was one of the goals of the project. The graphs in Figure 9 show the experimental discharge curves obtained for some of the batteries studied in the project. Four different lithium-ion cells were identified: an NMC pouch cell format with a nominal capacity of 210 Ah; a small cylindrical NCA cell (18,650 format) with a nominal capacity of 3.25 Ah; a larger cylindrical format LFP with an 85 Ah capacity; and an LTO prismatic cell with a capacity of 23 Ah. The typical flat voltage trend, in the central area of the LFP operating window of Figure 9a represents an optimal characteristic for both battery pack design and DC link coupling. In Figure 9b,d, it can be observed that the reduction in discharge capacity due to the increasing C-rate is extremely limited for NMC and LTO. This allows for good energy efficiencies at high C-rate discharges, despite LFP and NCA batteries showing a notable reduction in efficiency (Figure 9a,c). In particular, an energy loss of approximately 20% has been observed from 0.1C to 2C for NCA (Figure 9c) and an approximately 10% energy loss from 0.1C to 3C for LFP (Figure 9a). However, as can be seen in Figure 9, all of the analysed technologies were able to guarantee a high depth of discharge at various C-rates. Another detail in Figure 9 is the typical low operating voltage of LTO compared to other technologies (Figure 9d). This resulted in lower energy performance due to the presence of lithium titanate oxide in the anode. Nevertheless, although LTO batteries are not the best candidate for all-electric ships, they represent a good option for hybrid battery systems thanks to the ability to provide high-power pulses, a high number of cycles, and an impressive lifetime [40]. On the other hand, NMC and NCA batteries constitute the best compromise in terms of high specific energy and power, good durability, and a high number of cycles. LFP batteries represent the safest technologies due to their lower susceptibility to thermal runaway, but they have a lower specific energy and power than the previous technologies.

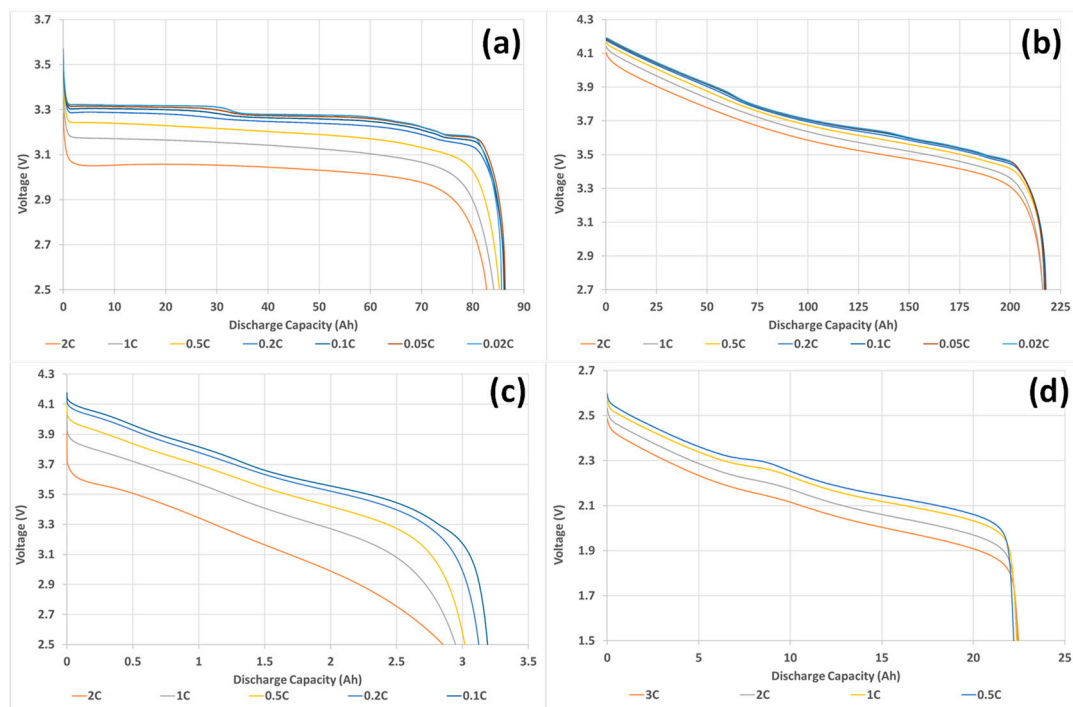


Figure 9. Experimental discharge curves at different C-rates for various cell chemistries: (a) LFP; (b) NMC; (c) NCA; (d) LTO.

Cost is one of the most important drivers in the battery selection process and is mainly influenced by electrode materials. They consist of highly complex and often rare metals, and many of these are critical raw materials [120]. Figure 10 displays the cost breakdown for the individual components that constitute a common lithium-ion battery cell. Materials represent the highest expense and account for about 65% of the total. The positive and negative electrodes are the most expensive parts of a battery. Their high cost is due to the use of large quantities of precious materials required in each cell. Labour is about 15% of the overall cost even though battery manufacturing is highly automated. The remaining 20% of the total cost includes cell and module assembly, and overhead costs due capital depreciation, energy costs, design, sales and distribution [48]. Depending on the battery's application (e.g., road or marine transportation), there are different costs to be considered for the safe integration of the battery according to applicable law [16]. For example, considering the same cell chemistry, for a marine battery system, the cost of the entire system can be about 40% higher than for a conventional automotive application [60].

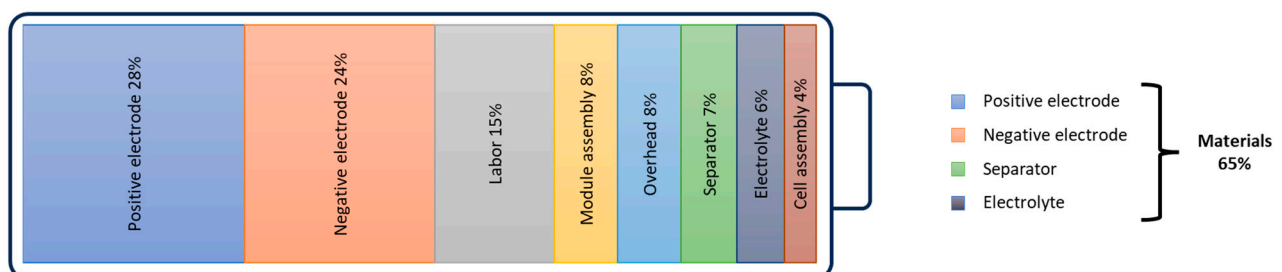


Figure 10. Cost breakdown by individual components, manufacturing, and overhead of a typical lithium-ion battery. Data from [48].

In general, the cost of a cell is just the starting point, not only because the main system consists of the assembly of battery packs, but also because the battery system must be integrated with auxiliaries. Therefore, to the cost of the modules and systems, it is necessary

to add the cost of different components such as the battery management system (BMS), power electronics, cables and connectors, internal cell housings and supports, additional components for control and security, and other components [55]. As an indication, for a LIB marine battery energy storage system, including power electronic, integration and installation, the cost in 2020 was in the range of 600 and 1000 euro/kWh, and this figure is expected to decrease by approximately 30% by 2030 and 50% by 2040 [16,121]. Additionally, when integrating large battery systems, it is important to consider the total capital costs in relation to the battery's end-of-life (EoL). Indeed, although the installation of a BESS increases system complexity and capital costs, it could be convenient considering the investment during the whole battery life. BESS profitability on ships has been proven for projects in which the high price of electricity from a battery is balanced by the high efficiency of electric propulsion [60]. The latter allows for direct savings due to the fuel consumption reduction and some indirect savings related to the minor ship maintenance that is required, thanks to a reduced number of engine running hours and the minor wear and tear of the components. Research studies comparing diesel–battery hybrid propulsion with a conventional diesel arrangement claim fuel savings in the order of 10–45% depending on ship type and operations [13,43]. Therefore, the total economic investment should be considered in relation with the aging performance of the selected BESS, and only this, would allow for the evaluation of a shipowner's return on investment (RoI).

4.2. From Cell to Battery System Configuration

The process of BESS integration in a ship can be divided into four phases, as per Figure 11. Considering the lithium-ion chemistries available on the market (discussed in Section 3.1), a trade-off among cost, energy, power, and aging performance will influence the selection of a commercial battery module based on the ship's energy profiles. Then, the battery pack is built by customizing the assembly of the modules, with series and parallel configurations, to reach the voltage and current values requested for the ship. The configuration of the battery system begins with the definition of the pack and its mechanical construction, together with the battery auxiliaries necessary for safe and functional operations such as power electronics; management systems; cooling and ventilation; detection; alarm; firefighting, and containment systems. After the BESS is installed, it will be possible to proceed with the electrical and physical integration of the BESS on the ship.

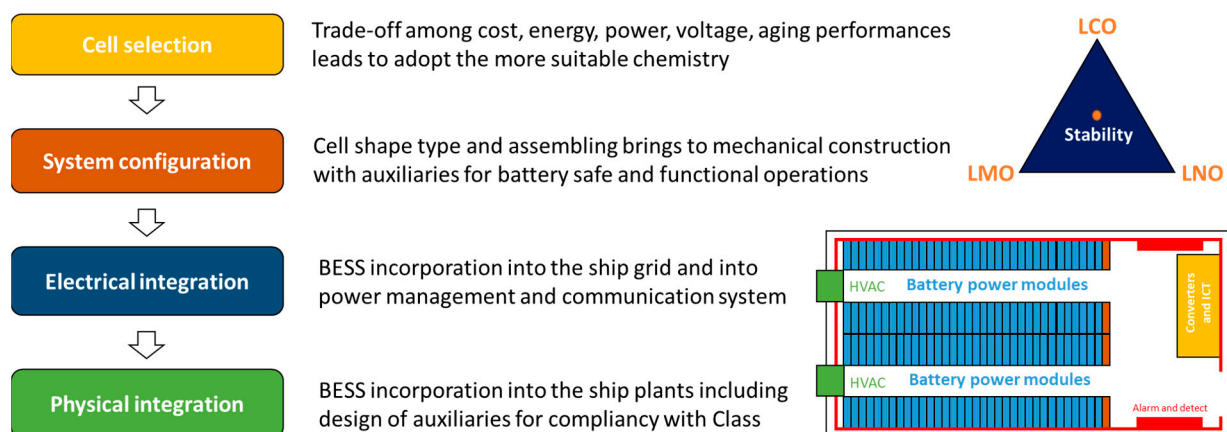


Figure 11. Battery energy storage system integration process.

4.2.1. Mechanical Construction and Casing

Cells are the primary functional unit of a battery, and they consist of electrodes, electrolyte, and terminals. These main components are assembled and suitably housed in an enclosed casing. There are four main types of cells depending on the manufacturing process and final shape. Coin cells are generally small and not suitable for making high-capacity modules or battery packs. Cylindrical cells are easy and economical, they usually

guarantee a long cell lifetime thanks to the geometry allowing for the easier cooling of the modules; however, this characteristic results in a low specific energy compared to prismatic and pouch cells. Pouch cells have the main advantage of having an inexpensive production process; they are also exceptionally light and are thinner than other types of cells, which gives flexibility in module design and allows for space optimization, resulting in good specific energy. On the other hand, they have lower mechanical strength, which requires the use of robust casings, thus increasing the weight of the module [122]. In all marine applications, single cells, modules, or packs are not sufficient to meet the energy, power, and voltage requirements; therefore, individual elements are connected in series and in parallel combinations to form large battery systems. Often, in addition to the electrical connections, marine battery systems also include electronic management devices (PCB/PCM, BMS), safety systems, and cooling systems within a waterproof housing [123] (see Figure 12).

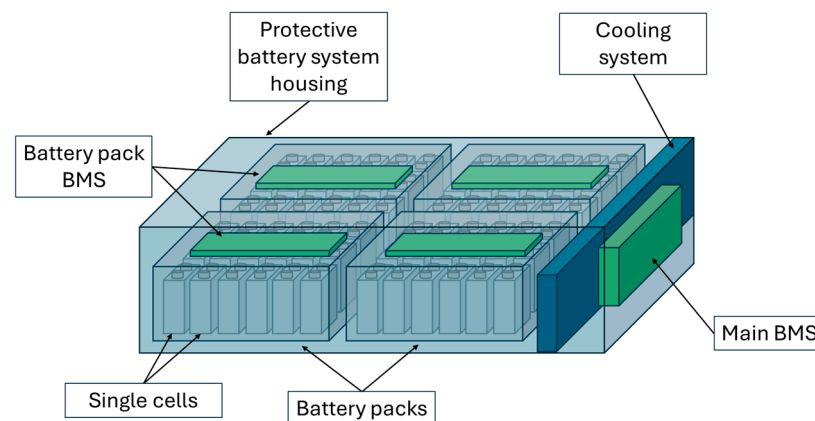


Figure 12. Battery energy storage system composition.

Adequate containment is always necessary to protect a battery from the marine environment and to preserve the batteries from harsh working conditions caused by navigation (e.g., vibration and vertical acceleration). Consequently, marine battery packs must be placed in a corrosion-resistant enclosure along with sealed marine-grade connectors. The precise layout of the battery casing must be designed, on a case-by-case basis, based on the available space on board and the specific ship systems to be interconnected with the BESS [116]. Depending on the application, commercial solutions offering containerized battery systems can also be used, including all auxiliary systems for the safe and functional operation of the batteries (energy management system (EMS), cooling system, firefighting system, detection and alarm systems, power converters, and all related equipment).

4.2.2. Considerations at System Level

As discussed in Section 4.1, safety is one of the key factors when selecting batteries for ships. Firstly, because of unacceptable consequences in terms of human risk and economic losses resulting from accidents, and secondly, because large BESS applications on board ships are relatively recent and, therefore, there are no sufficient sailing test cases yet. For this reason, current BESS designs have a very high level of redundancy, resulting in little optimization in favour of the necessary high level of safety. In practical terms, this means that all the cell components must be designed and tested to ensure elevated standards (e.g., resistance to fire and chemical risks). The risks deriving from the use of batteries can depend on causes internal to the cell, such as manufacturing defects and consequent self-heating, or on external causes, such as electrical, mechanical, or thermal abuse [124].

The safety level of a cell varies depending on its components. For instance, lead-acid batteries are stable and safe, with chemical risks primarily related to the release of toxic substances under external conditions. LIB batteries, on the other hand, present fire risks caused by the presence of flammable metals and oxidizing agents. Thermal runaway is a well-known phenomenon in LIB batteries, which can arise from various causes (internal or

external short circuits, operation beyond permitted limits, degradation phenomena, etc.) and must be avoided. Risks from internal causes can be mitigated with proper battery selection; some of the possible variables affecting safety are cell materials and mechanical construction. Risks from external causes, however, are more complex to address in the design phase, because they are often linked to the specific use of the batteries and the integration of the BESS with other ship systems such as HVAC, Fire Prevention, and Automation.

At the system level, it is necessary to consider the combination of the individual effects discussed at the cell level in Section 3.3, and add other hazards resulting from cell scale-up in modules composed of thousands of interconnected cells. Therefore, during the integration process of a BESS, numerous technical considerations on mechanical, chemical, electrical, and fire phenomena are required. In addition, to evaluate the safety and functionality of the entire battery system, it is necessary to consider the operational modes of the vessel that could cause safety hazards related to electrical operation.

In addition to battery-specific electrical, thermal, and other risks, when evaluating onboard BESS installations, there are several aspects that require a risk-based design approach. Therefore, to design and install a suitable BESS on a ship, it is necessary for a risk assessment to consider all possible hazards. After having assessed which risks are acceptable, consequent technical and operational countermeasures are adopted (e.g., active and passive firefighting and containment systems, special ventilation, dedicated escape routes, etc.) [12].

The monitoring and mitigation of operational risks, (e.g., overcharge, over-discharge, overcurrent, overheating, short circuit) are always necessary and can be achieved through battery and vessel protection systems. The battery management system (BMS) controls and monitors the operational status of the battery at various levels (cell, module, and pack). The BMS helps to maximize battery life, efficiency, and safety. Among the fundamental functions of the BMS, of primary importance is the protection against short circuits carried out by maintaining voltage and current within the safety limits imposed by the manufacturer. Using the BMS to maximize battery life is achieved with algorithms that provide accurate estimates of the SoH and SoC to the energy management system/unit (EMS). Other specific tasks performed by the BMS are thermal management and charge control: the first for battery protection and safety control, the second for cell monitoring and balancing [125]. As shown in Figure 13, a typical marine BMS can engage in the real-time monitoring and control of safety and performance parameters such as temperature, voltage, SoH, SoC, DoD, current, and cooling level [116].

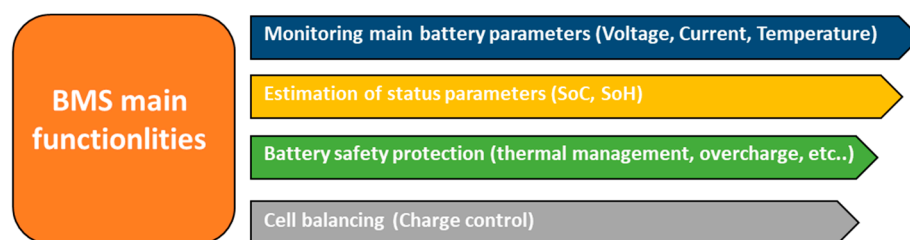


Figure 13. Key battery management system features for ship batteries.

In terms of hardware and software, in a BMS, there are one or more microcontrollers, integrated with appropriately developed firmware codes, capable of monitoring the entire battery system and, when necessary, carrying out actions such as cell balancing. This last function is particularly important to prevent the uncontrolled aging of the different cells or to avoid the non-optimal charging and discharging of the pack. Passive balancing dissipates energy by creating waste heat and, for this reason, is not efficient in terms of energy management. Active balancing, on the other hand, shifts energy from cells with a higher SoC to cells with a lower SoC. A less expensive alternative to active balancing, more common for large battery packs used in the marine industry, is to disconnect the

charged cells when they are fully charged. Battery protection is achieved with a protection circuit for small battery packs; however, for high-voltage batteries, the BMS structure can be more sophisticated, incorporating dedicated modules for parameter control, display, calculation, data conversion (analogue to digital conversion), and hosting units for data collection, management, and communication.

4.3. Battery Electrical and ITC Integration

The integration of a battery system into a ship electric power system can be realized with different approaches, depending on the different ship electric grids (Direct or Alternate Current main distribution) and their topologies. After connecting the BESS to the ship network, incorporating into the power management system and integrating with the communication system are necessary.

4.3.1. Ship Electrical Grid Topology and Distribution

The ship's grid topology depends mainly on the type of propulsive system (mechanical, hybrid, or electric). Conventionally, ships have separate electrical power sources dedicated for propulsion, auxiliary systems, and emergency purposes. Propulsion power is produced by internal combustion engines (main engines); power for auxiliaries and for emergencies is instead produced by generator sets consisting of an electrical generator driven by an internal combustion engine (generator sets) [126]. Traditionally, ships have been mechanically propelled by a main engine connected to a shaft with a gearbox, providing torque to the propellers. However, since the late 1980s, electric propulsion has been introduced to increase propulsive and energy efficiency, to have more comfort (reduction of noise and vibrations), to reduce maintenance costs, and to optimize onboard space utilization [6]. Therefore, due to fluctuating ship power demands (e.g., offshore supply vessels) or specific space requirements (e.g., cruise ships), currently there is a wide range of electric propulsion trains with propellers driven by electrical motors (EMs). Typical arrangements comparing conventional ship propulsion trains are shown in Figure 14 where, in the block diagrams, EMs are connected to the ship's main switchboard via converters and all of the necessary electrical equipment. Propulsive energy generation in Figure 14a is provided by the main engine, instead, in Figure 14b,c, it is provided by dedicated generator sets (segregated electric propulsion) or the ship's unique power generation source used either for propulsion or for auxiliary loads (integrated electric propulsion) [1]. Electric motors can be located close to the propeller inside the ship or outside of the ship, inside dedicated gondolas (podded propulsion) [6]. There are also hybrid propulsion trains consisting of two parallel propulsion systems: one mechanically driven from the main engine, and the other electrically driven by a motor, able to work independently or in combination [22].

Depending on the ship's grid topology, it is possible to integrate batteries for propulsion or for ship energy services (see Section 2). When integrating a battery system, one of the first tasks is to design the distribution system considering the desired battery location in the ship's grid. Placing a BESS before or after the main switchboards can influence the efficiency and redundancy of the whole propulsion system. For example, in electrically propelled offshore vessels (DP class), it is possible to connect distributed batteries directly into the propulsion converters, as shown in the block diagrams of Figure 15a. Such a grid, compared with conventional battery-electrical propulsion (Figure 15b), has increased electrical efficiency and higher redundancy during manoeuvring thanks to independent propulsion units with their own source of energy [126].

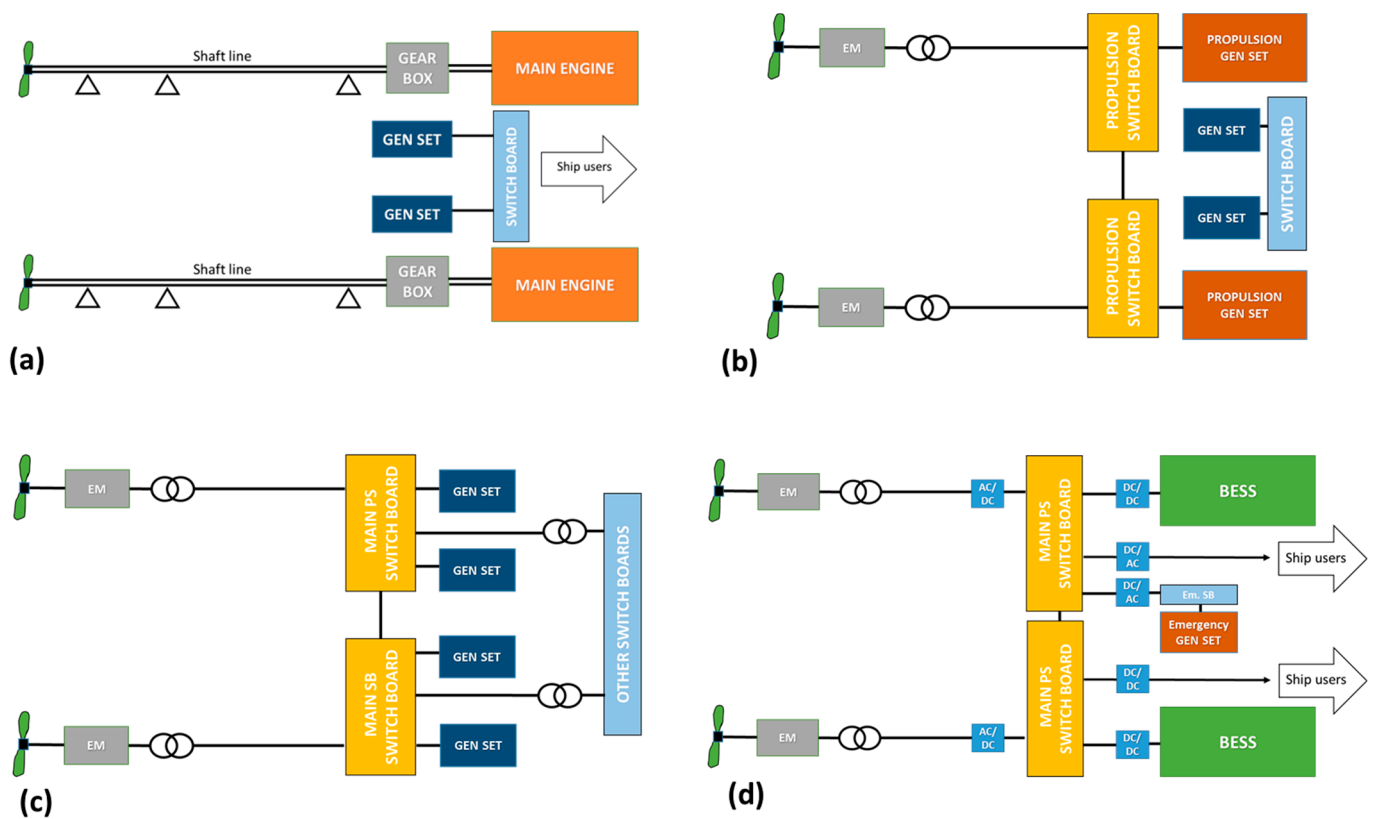


Figure 14. Different ship propulsion trains: (a) mechanical; (b) electrical segregated; (c) electrical integrated; (d) all electric ship.

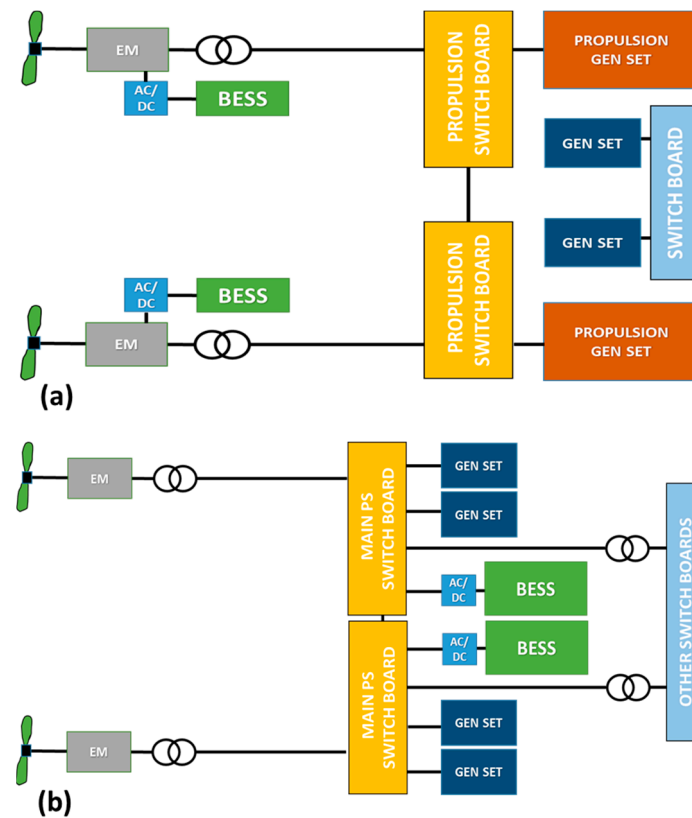


Figure 15. Two possible topologies for ship electrical propulsion with batteries: (a) distributed batteries; (b) conventional battery electrical propulsion.

Ship electrical distribution systems have evolved considerably in the last century and today, a ship network is made-up of a complex architecture, which must interface with various loads and power sources with variable power and voltage levels. For this reason, modern ship electrical power systems become more or less similar to terrestrial microgrids [127] and the technologies developed for island microgrids can be extended to ship microgrids [128]. Traditional ship power distribution systems are based on an alternating current (AC) grid at low voltage (LV). Due to the increase in power demand, today, AC medium-voltage grids (3.3 to 13.8 kV) are also widespread, especially on large ships. AC distribution systems require a fixed frequency and, therefore, generators must run at a fixed speed under variable loads. This does not always guarantee optimal operation and, in AC networks, it requires the synchronization of the various generators. These limitations are just some of the factors that have favoured the distribution of medium voltage direct current (DC) networks from 1 kV to up to 35 kV on ships [129]. Furthermore, a DC distribution system, compared to AC systems, allows for weight savings thanks to the possibility of eliminating some transformers (e.g., low-frequency transformers) and gives more freedom in selecting electrical machines for energy generation (e.g., high-speed machines with low weight) [130]. Some other benefits of DC distribution on ships are the ability to have a more easily controlled and bi-directional power flow, the ready integration of storage systems such as BESSs and other electrical components, the possibility of working at various frequencies, and the absence of harmonic distortions in the network [131].

4.3.2. BESSs for Power Quality

Compared to islanded terrestrial grids, ship electrical power grids are weaker due to the complex power electronics that are used and due to the presence of single loads with powers in the same order of magnitude as the total installed power. On the one hand, the probability and magnitude of perturbations occurring in ship grids are greater and the overall requirements are more stringent due to both reliability and safety targets imposed by marine rules and regulations (see Section 5) [6]. Nevertheless, most power quality issues can be attributed to voltage variations (AC and DC microgrids), frequency variations (AC microgrids), and harmonic waveform distortions which are due to cyclic or non-cyclic load transients in AC ship microgrids.

Improving the quality of ship grids is possible using the experience from islanded terrestrial microgrids, with the main difference being that on ships it is not possible to oversize systems due to weight and space limitations. BESSs for ship microgrids are DC in nature and normally operate at low voltage levels. Thus, interfacing them with other microgrid elements, through DC–DC or DC–AC converters, is essential for integrating them with ship power systems. For ships, it is common to use energy storage systems with inverter-interfaced generation sources and in such networks, the output current can be controlled instantaneously to mitigate power quality issues. Nevertheless, it should be noted that the rapid response capability of inverter systems is limited by the characteristics of the energy storage devices, that need to have a high specific energy, and the capability to deliver energy as fast as the inverter needs.

The main power electronic converters on ships are used for generators, propulsion motor drives, thrusters, pumps, and fans. Due to the high-power level and the relatively low frequency, they introduce harmonic waveform distortions. These can be mitigated with passive filters (inductors and capacitors) or with active filters. In the latter solution, a BESS is interfaced with the microgrid through bidirectional power converters, able to control both active and reactive power instantaneously and to mitigate harmonics produced by large motor drives. For example, in low-voltage AC ship grids that are more resistive than inductive, the active power has a greater influence on the voltage, so a BESS can be used to exchange active power for voltage regulation with reactive power exchange being controlled to regulate the frequency. Commonly, in more inductive and high voltage systems, the opposite approach is used. In DC ship power systems, the control of the interfacing power converter does not involve synchronization of frequency control, hence

BESSs are used in a more straightforward manner to regulate the voltage by providing or absorbing the deficient or surplus power [131].

4.3.3. Integrated Power and Energy System

An integrated electrical power system (IEPS) is required on board a ship to provide power generation, distribution, and control due to the various elements that need to be coordinated. BESSs are therefore not excluded from this integration discussion, but they represent a core element of the power chain. Historically, the integrated power system (IPS) allowed more controllability in AC ship propulsion systems, and thus enabled the use of a common power system for both propulsion and service loads. Today, the IEPS is a key component in electrically propelled ships for efficiency and safety reasons. Indeed, due to the typical ship propulsion system dynamics, it significantly impacts the ship's microgrid power quality [131]. On board, an IEPS must satisfy requirements that are different from stationary terrestrial power systems. The first difference concerns the distinction between essential and nonessential users. Essential users are composed of power loads, whose supply and correct service must always be assured, also in the case of a major system fault (defined by regulations). Their functionalities are crucial for the safe operation of the ship's propulsion and manoeuvring systems, fire suppression, communication, emergency, and navigation systems. A second major concern is related to the need for the functional integration of very different systems such as: a large power station with generators working at high voltages (HV, above 1 kV); a main HV distribution system; a secondary distribution system (LV); a large amount of electrical machinery of different types, both HV and LV, with either direct or variable speed drives. A third difference is that a modern ship IEPS exploits the extensive use of power electronics and automation, such as real-time control systems (lower automation layers), and distributed automation systems (higher automation layers) [6]. Moreover, the complexity of the tasks performed by the IEPS is high because, depending on functionalities, the priorities are different for each considered shipload. Therefore, control strategies for the coordination of different loads with a differentiation of their responses is an essential feature in modern marine power systems. As a result, a modern ship electrical distribution and power system requires a comprehensive understanding of load profiles and complex communication and coordination strategies [6]. Finally, advanced fault detection, identification, and isolation algorithms are essential for the successful implementation of zonal electrical distribution systems in ship microgrids that become necessary in order to ensure higher survivability, reliability, and improved efficiency [132].

Therefore, integrating BESSs with conventional ship electrical architectures is a task with a high level of complexity. The absence of a tie line connection makes ship power systems (SPS) vulnerable to failures [133], and the proper use of the BESS can be crucial to reach higher resilience and more efficient power management. Some studies faced the technical aspects related to grid management with the ship energy management system (EMS). In such architecture, the BMS of the battery is interfaced with the EMS and a power management system (PMS). The EMS constitutes the whole-ship system supervisor, while the PMS maintains the overall control of electrical power production and consumption [13]. Furthermore, with the objective of managing the ship's energy flow, different power splitting techniques derived from the automotive industry can be used, such as dynamic programming (DP), equivalent fuel consumption minimization strategies (ECMS), or model predictive control (MPC) as reported by Planakis et al. [134]. Here, a new framework for energy management that implements machine learning, data driven techniques, and optimal control is proposed. More specifically, an integrated, real-time NMPC (non-linear MPC)-based energy management control system for parallel hybrid diesel–electric propulsion plants is developed and evaluated. Ship energy management, as a multi-objective optimization problem, is faced in [133] where, after a linearization process, it is solved using the mixed-integer linear programming (MILP) technique. Another bi-level multi-objective differential evolution algorithm is reported in [135]; in this work

some defined parameters, related to navigation and the systems involved, are optimized at a higher level while specific power generation scheduling is settled at a lower level.

In all cases, today, energy management on ships remains an open issue. Various design techniques and approaches (software simulators, hardware in the loop, etc.) are already in place to face the challenge [6]. Technological research on these kinds of systems is shifting from the IEPS to the integrated electrical and electronic power systems (IEEPS) concept. The only way to achieve the required features in complex power architectures, such as those present on ships, seems to be the extensive adoption of power electronics, highly integrated into single subsystems [6]. This is possible thanks to the deployment of modern information and communications technologies (ICT) allowing for the correct coding of routines and the efficient transfer and management of the data necessary to provide input and output to all the programmed ship automation processes. Therefore, in order to build an efficient and safe automation and control system on board a ship with BESS, it is necessary to use modern communication technologies, such as controller area network (CAN), local area network (LAN), protection algorithms, and complex monitoring systems (e.g., multi-functional monitoring systems and decision-making algorithms) [131,136].

4.4. Physical Battery Integration on Board

Ship battery dimensions can be a limiting design factor; indeed, onboard space is restricted and only a limited amount of weight can be carried on the ship to ensure regulatory flooding limits and stability performance. Thus, marine battery modules are assembled with more packs connected properly in series and in parallel to optimize weight and volume while reaching the desired voltage and energy. Module dimensions depend on the type of cell used for assembly. Usually, large cells have the advantage of minimizing the materials used for the construction of the external casing. This translates into a higher specific energy when used to make a battery module. Furthermore, due to their larger dimensions, they are also more robust and mechanically stable. For this reason and for easier integration, unless there are specific high-power requests, big cells are to be preferred. In fact, with a larger cell size, it is possible to reach the required energy capacity with a smaller number of cells, and this helps save space, more easily achieve cell electrical balancing, and reduce the risk of failure. On the other hand, the main advantage of small cells is the possibility of reaching higher specific powers and the possibility of mitigating the consequences of failure if properly insulated.

Physical integration is not limited to just finding space on board, but also involves the modification of conventional ship systems that need to be adapted to provide battery services. The ventilation and air conditioning system ensures a constant temperature and humidity in the battery room. Passive fire protection of the battery is realized with the use of fire-resistant materials to contain possible flames. The installation of fire, smoke, and temperature sensors is necessary to detect hazards, and proper firefighting systems are installed to extinguish fires in a timely manner. Extra pipes, ducts, and cables must be routed from ship systems to ensure the transmission and conversion of electricity and to serve conventional and emergency systems. In addition, the location of the battery space must be selected to satisfy ship statutory and Class rules. Locations in the lower part of the ship and, as much as possible, far from sides of the ship are to be preferred for minimizing the consequences in terms of stability and survivability. Once a battery room is designed, according to applicable Class rules prescription [126], due to the relevant weight addition, an assessment of ship deadweight and ship stability in intact and damaged conditions must be performed, to confirm that the selected battery system may be carried on board.

5. Regulations for BESSs: Class Rules and International Codes

Battery use on ships involve national and international regulatory aspects concerning environmental issues, as regulated by “MARPOL Annex VI”. In that respect, since 2011, the International Maritime Organization (IMO) is pursuing several actions with the goal of improving the sustainability of the shipping industry. The European Union (EU) plays a

leading role in promoting the reduction of CO₂ emissions from maritime transport, issuing regulations on the monitoring, reporting, and verification of CO₂ emissions from ships accessing ports under the jurisdiction of an EU member state [137].

The regulatory framework for ships is complex, since all marine businesses related to the design, construction, dismantling, and operation of ships are regulated by non-governmental organizations, such as ship classification societies. In addition, it is necessary to comply with international regulations from the International Maritime Organization (IMO), national regulations (flag states), and technical standards (IEAC, UL, ISO). Therefore, during battery system design and integration, it is necessary to consider the general requirements of a ship in different matters, including electric equipment, the generation and use of energy on board, the management of dangerous materials, and other matters.

5.1. Approval Process for BESSs on Board Ships and Technical Standards

Considering the novelty of BESSs on board ships, during design and installation, it is not possible to apply a prescriptive approach, due to the issue not yet being fully covered by existing rules. In addition, different existing ship types and the variety of possible battery services (see Section 2) make it difficult to find universal rules that are applicable to all cases. Therefore, the battery design and integration process has to follow a goal-based approach, and every design requires a tailor-made solution [138].

The design and approval process of a battery system involves many players (IMO, flags, classification societies, shipowners, ship operators, and various manufacturers) and, with BESS installations on board ships being a novel design feature, these players might not be perfectly aligned in terms of guidelines and regulations. Thus, to ensure a smooth Class approval process, all the misalignments must be removed thanks to a push from the involved stakeholders. Classification societies structure this process, which is facilitated by technical intermediate phases such as: type approval, approval in principle, risk-based design, flag circulars, guidelines, and formal safety assessments.

Among the international codes and technical standards for the installation of batteries selected in Table 3, some specific standards are referred to in similar applications in the stationary and civil industry, such as UN 38.3, IEC 60529, IEC 60092-504, IEC 62061, IEC 61508, IEC 62619, IEC 62620, UL 1642, UL 1973, and UL 9540.

Table 3. Non-exhaustive list of the main technical codes and standards for the installation of batteries.

Date	Standard/Code	Title	Note	Ref
2022	IEC 62619:2022 CMV	Secondary cells and batteries containing alkaline or other non-acid electrolytes—Safety requirements for secondary lithium cells and batteries, for use in industrial applications		[139]
2023	IEC 62620:2014/ AMD1:2023	Amendment 1—Secondary cells and batteries containing alkaline or other non-acid electrolytes—Secondary lithium cells and batteries for use in industrial applications		[140]
2022	UL 1973	Batteries for Use in Stationary and Motive Auxiliary Power Applications		[141]
2020	UL 1642	Standard for Safety of Lithium Batteries		[142]
2019	UN 38.3	Manual of Tests and Criteria on Transport of Dangerous Goods		[143]
2019	IEC 62281: 2019+AMD1:2021+ AMD2:2023 CSV	Safety of primary and secondary lithium cells and batteries during transport		[144]
2021	UL 9540	Energy Storage Systems and Equipment		[145]
2013	IEC 60529: 1989+AMD1:1999+ AMD2:2013 CSV	Degrees of protection provided by enclosures (IP Code)		[146]

Table 3. *Cont.*

Date	Standard/Code	Title	Note	Ref
2010	IEC 61508:2010 CMV	Functional safety of electrical/electronic/programmable electronic safety-related systems		[147]
2016	IEC 60092-504	Electrical installations in ships—Part 504: Automation, control and instrumentation	Relevant for BMS	[148]
2021	IEC 62061:2021	Safety of machinery—Functional safety of safety-related control systems		[149]
2023	EN 50110	Operation of electrical installations—Part 1: General requirements	Supporting electrical testing of batteries for ships	[150]
2002	IEEE 45-2002	Recommended Practice for Electrical Installations on Shipboard		[151]
2012	IEEE 80005-1	Utility connections in port—Part 1: High Voltage Shore Connection (HVSC) Systems—General requirements		[152]
2016	IEEE 80005-2	Utility connections in port—Part 2: High and low voltage shore connection systems—Data communication for monitoring and control		[153]
2014	IEC PAS 80005-3:2014	Utility connections in port—Part 3: Low Voltage Shore Connection (LVSC) Systems—General requirements		[154]
2019	IEC 62613-1:2019	Plugs, socket-outlets and ship couplers for high-voltage shore connection (HVSC) systems—Part 1: General requirements	Relevant standards for land to ship connections relating to the use of shore power	[155]
2016	IEC 62613-2:2016	Plugs, socket-outlets and ship couplers for high-voltage shore connection systems (HVSC-systems)—Part 2: Dimensional compatibility and interchangeability requirements for accessories to be used by various types of ships		[156]
2017	IEC 60309-5:2017	Plugs, socket-outlets and couplers for industrial purposes—Part 5: Dimensional compatibility and interchangeability requirements for plugs, socket-outlets, ship connectors and ship inlets for low-voltage shore connection systems (LVSC)		[157]

The IMO “International Convention for the Safety of Life at Sea” (SOLAS) defines the minimum internationally agreed requirements for ship construction, equipment, and operation. More specifically, in the section regarding electrical power generation, some requirements that are also applicable to BESSs are provided (See “SOLAS—Chapter II-1—Construction—Subdivision and stability, machinery and electrical installations”). IMO “MSC Circ. 1455” dated 2013 (“Guidelines for the approval of alternatives and equivalents as provided for in various IMO instruments”) describes the approval process for novel technologies and it is considered a reference for safety and technical requirements in the absence of relevant Class and flag rules. The IMDG Code “International Maritime Dangerous Goods Code” instead covers the transportation of dangerous cargo goods and, to some extent, is applicable since lithium batteries are composed of dangerous substances.

5.2. General Technical Content of BESS Rules for Ship and Comparison among Main Class Societies

Class rules contain prescriptions concerning mechanical, HVAC, electrical, COM-IT, and safety aspects. Primarily, there are indications to ensure proper battery mechanical integrity and resistance to the harsh marine environment.

A non-exhaustive summary of main applicable Class rules for BESSs is proposed in Table 4. The classification societies that are analysed and compared are “Det Norske Veritas” (DNV), “American Bureau of Shipping” (ABS), “Bureau Veritas” (BV), “Registro Navale Italiano” (RINA), and “Lloyd Register” (LR), because these (all IACS members) are the

pioneers in regulating and supporting the application of emerging technologies, from the embryonal phase to the definition of official guidelines and regulations.

Table 4. List of main Class rules applicable to the installation of batteries on board ships.

Class	Type of Document	Title
DNV	Class rule	DNV-GL General Rules Part 6 Additional Class Notation, Ch 2, Section 1
DNV	Guideline	DNV-GL Handbook for Maritime and Offshore Battery Systems
DNV	Technical code	DNV-GL CP 0418 Electrical Energy Storage
DNV	Class rule	DNV GL rules for classification—ShipsRU SHIP Pt.4 Ch.8—Electrical Installations
DNV	Class rule	DNV GL rules for classification—ShipsRU SHIP Pt.4 Ch.9—Control and monitoring systems
DNV	Class rule	DNV GL rules for classification—ShipsRU SHIP Pt.6 Ch.2 Sec.1—Battery power
DNV	Technical Standard	DNV GL offshore standard OS D201 Electrical installations
DNV	Technical Standard	DNV GL offshore standard OS D202 Automation, Safety and Telecommunication Systems
DNV	Guideline	Class guideline DNV GL CG 0339—Environmental test specification for electrical, electronic and programmable equipment and systems
DNV	Guideline	DNV GL Guideline for large Maritime Battery Systems
ABS	Guideline	Use of lithium-ion Batteries in the Marine and Offshore Industries
ABS	Class rule	ABS Rules for Building and Classing Marine Vessels PART 4
ABS	Class rule	ABS Advisory on Hybrid Electric Power Systems
ABS	Class rule	ABS Rules for Building and Classing Marine Vessels PART 1 Ch 1 Sec 4/ Alternatives
ABS	Class rule	ABS Rules for Building and Classing Marine Vessels PART 1 Ch 1 Appendix 3/ ABS Type Approval program
ABS	Class rule	ABS Rules for Building and Classing Marine Vessels PART 1 Ch 1 Appendix 4/ ABS Type Approval program
LR	Guideline	Large Battery Installations: Key hazards to consider and Lloyd’s Register approach to approval
RINA	Class rule	Rules for the Certification, Installation and Testing of Lithium-Based Storage Batteries
ABS	Guideline	Guide for use of lithium batteries in the marine and offshore industries
BV	Class rule	Rules for Classification of Ships—Electric Hybrid (Pt F, Ch 11, Sec 22)

All analysed battery Class rules refer to system performances and they do not prescribe differences for specific chemistries. They provide guidance and limitations that are applicable to big electrical installations, with special regard to cases of batteries used for propulsion. Being that these regulations are very extensive documents, covering several topics, there are prescriptions in many sections. Therefore, the applicable rules can be found in the energy production and fuel storage sections. Safety prescriptions are often included in dedicated technical codes prescribing limitations to guarantee the proper level of safety that is necessary for the integrity of the ship and its cargo or passengers.

General topics that are covered related to safety are: the potential exhaust of harmful gases (toxic, flammable, corrosive); the risk of fire and requirements for fire-fighting system; the risk of explosion; the risk in case of internal and external short circuits; and the required electrical protections [158,159]. All Classes prescribe requirements regarding the use of fireproof materials for the construction of the external battery case. Usually, a minimum ingress protection level of IP44 is required, meaning protection against solid bodies larger than 1 mm and against water splashes [126]. Other rules prescribe that the battery system must be equipped with an emergency shutdown mechanism outside of the battery space and that the emergency shut-off circuit must be wired and made independent of any control circuits. The battery system must be capable of being electrically isolated for maintenance purposes. The casing of each individual battery module must be equipped with a vent device to avoid rupture or explosion. The individual modules must be prepared in a way

that prevents the leakage of the electrolyte. All battery system output circuits must be protected against overloads and short circuits.

Concerning the necessary battery auxiliary space together with the relative systems, the BV Class [160] provides some specific requirements about the following: ventilation air changes per battery room; insulation against liquids or solid objects entering into the battery compartment and BMS casing; protection against electrostatic discharges using suitable antistatic paints; provision of sensors for gas and fire detection; provision of fire extinguishing systems; provision of safe entrances for accessing and maintaining the battery system. The RINA Class [161] provides indications on the required automation for valves that are necessary to release possible overpressure inside the battery casing.

Focusing on the battery management system, the DNV GL provides indications concerning the minimum requirements for the BMS that, in all cases, needs to provide for the balancing of the cells and the monitoring of the temperature, voltage, current, SoH, and SoC. Furthermore, according to the DNV GL, the BMS must have the ability to control and timely protect the BESS against overcurrent, overvoltage, and overheating.

Regarding the tests to be conducted at the cell and module level, references are made to the IEC 62619 and IEC 62620 standards prescribing functional and safety tests for lithium batteries such as: external short circuit; impact; thermal abuse; overcharge; forced discharge; thermal runaway; failure of any sensors; cell balancing, and capacity assessment. At the system level, there are tests to be conducted to prove resistance against vibration, heat, humidity, temperature, corrosion, and flammability [162].

Specifically for the battery systems that are used for propulsion, some classification societies [163] provide additional requirements prescribing the redundancy of at least two independent battery systems located in separate spaces; minimum capacity according to the operative profile and battery service; the presence of an adequate power management system (PMS); additional protection circuits; and a load shedding function.

6. Conclusions

Despite maritime traffic producing a limited amount of emissions compared to other industries, the global climate crisis has led to changes in market and regulatory requirements, prompting a sensible reduction in emissions also in the maritime industry. The electrification of propulsion and onboard ship systems is one way forward to meet emission requirements while improving overall ship efficiency. This is possible with the integration of BESSs for fully electric and hybrid ships.

In this work, batteries have been proposed as an enabler of the energy transition in shipping. They are deployed both for energy production and energy management, since BESSs can work either as recipients for alternative energy sources or as energy buffers for energy management actions. Most of the lithium-ion battery chemistries that are available on the market have been investigated in this review in regard to marine applications, considering their technical and economical pros and cons.

BESS applications for ships can be beneficial in terms of energy efficiency and for mitigating ship operating costs. These benefits can be realized through the implementation of several electrical services, such as spinning reserve, load regulation, peak shaving, energy harvesting, regenerative power, ramp levelling, etc. These services must be selected, case-by-case, depending on different ship types and specific mission profiles, as illustrated in this review.

Currently, the marine application of BESSs requires the adoption of lithium-ion technologies, which are the current trade-off solution that is able to offer good efficiency for many ship services but is still not at the level of conventional diesel engines and generators. Despite this, fully electric battery propulsion can already be cost-effective for some specific ship types, such as RoRo, tugs, and high-speed crafts. Moreover, observing the specific energy values expected from batteries under research and development (including solid-state and metal-air batteries), a wider range of full electric ships would be possible whenever these technologies become available and allow for sailing autonomies more

comparable with current commercial mission profiles. Emerging sodium-ion batteries are gaining traction as a promising alternative to lithium-ion technology, also in shipping industry. This technology shows great potential for cost savings and for providing improved safety features that are essential for ships. Indeed, the widespread adoption of LIBs faces obstacles due to safety challenges associated with onboard use coupled with a relatively limited project track record in ship installations. Therefore, waiting for more validations to come from the field, current lithium-ion battery implementation is subject to elevated redundancy requirements imposed by strict rules from international bodies and regulated by classification societies.

The integration of batteries in ship electrical grids requires the analysis of the propulsive arrangements and the modification of conventional energy management on board. Despite the traditional AC ship grid, today, several other architectures are possible. In fact, depending on the application, a BESS can be adopted for either DC or AC networks, not only for propulsion and other ship electrical services, but also for improving ship grid power quality. In this regard, experience from terrestrial island microgrids is often extended to ship microgrids.

The installation of battery systems and their auxiliaries must be arranged to guarantee high survivability and reliability while maximizing ship efficiency. Therefore, physical and electrical integration has to come with the optimization of ship energy use through a complex integrated power and energy system (IPES) that is able to connect, monitor, and control many onboard plants such as ventilation, cooling, detecting, alarm, and firefighting. Consequently, a large battery installation on ships requires a modern ship automation system that can interconnect different systems to allow specific electrical loads in accordance with variable ship priorities as defined by operational and safety requirements.

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